Research Methods

INFO B302 – Introduction à la démarche scientifique (IDS)

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What is: the State-of-the-art

…things that were… things that are…

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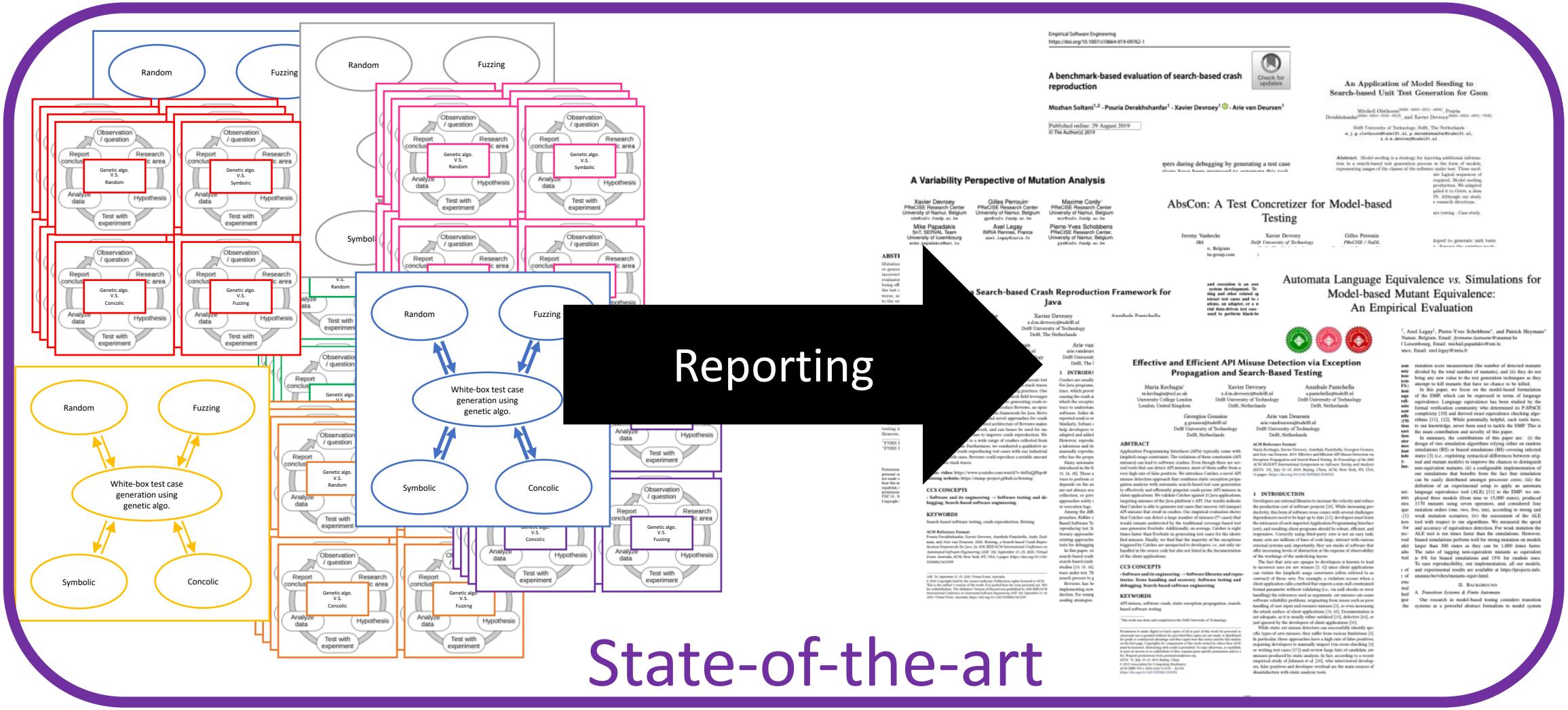
Reporting in (peer-reviewed) publications



Systematic Literature Review

1. Plan the review

1. Specify research questions
2. Develop review protocol
3. Validate review protocol

2. Conduct the review

1. Identify relevant research
2. Select primary studie
3. Assess study quality
4. Extract required data
5. Synthesise data

3. Document the review

1. Write review report
2. Validate report

What could be: Research questions

and some things…that have not yet come to pass.

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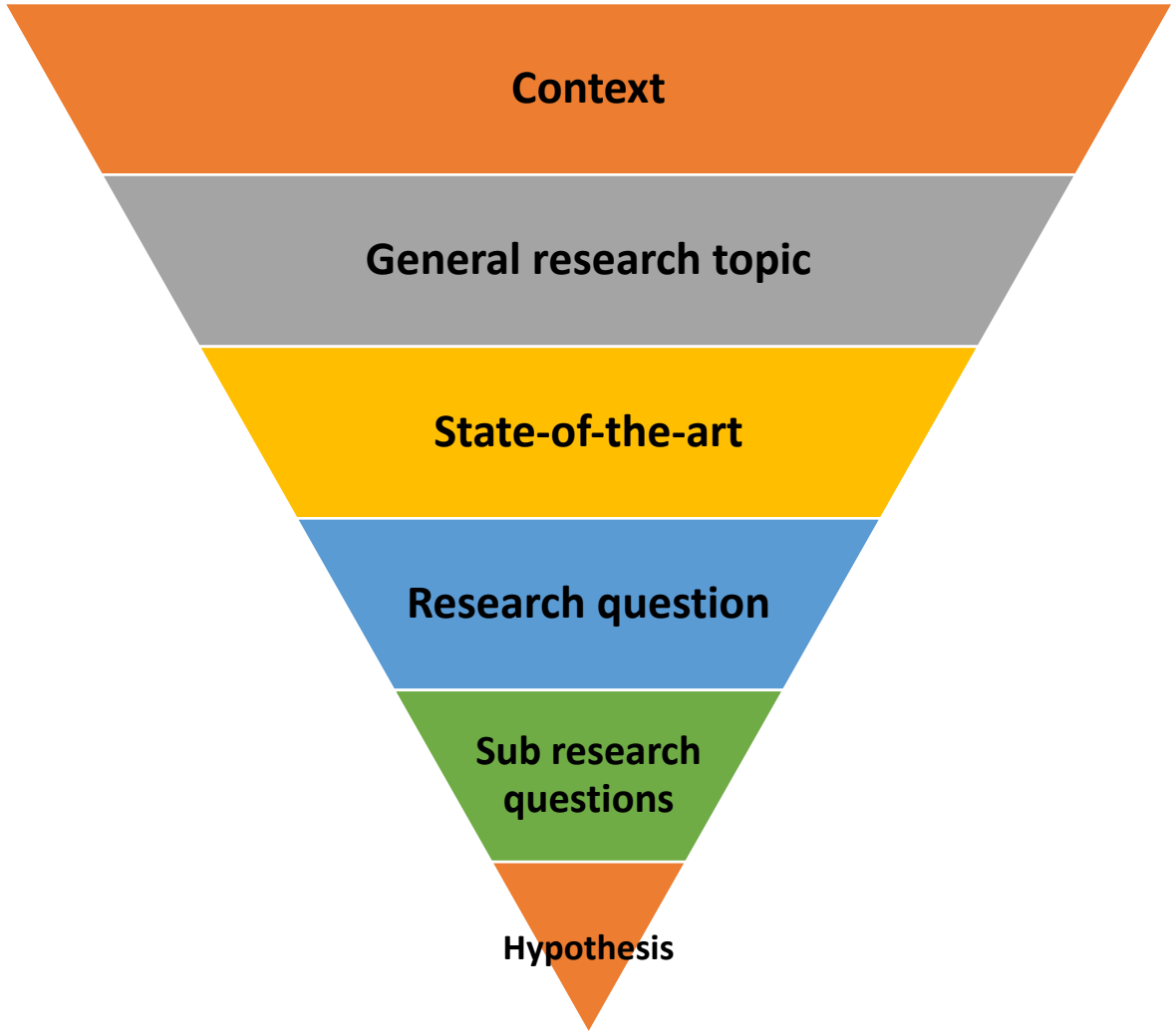
Building a research question

- Context and general topic
- Check the state-of-the-art
- Narrow down to a specific niche to focus on
- Identify a research problem to address
- Formulate the research question



Research method

Let’s check with data!

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Research question

Why?

How does the approach XYZ **compares** to A, B, and C regarding its **fault-finding** capabilities in **Java programs** with time budgets of 1, 3, and 5 minutes?

Measure?

On what?

How? 🤷

Research method

Maximize confidence in the findings

Decrease the threats to the validity

- What *confidence*?
- What *threats*?
- How?

- … it depends 🤷

Example

A. Panichella, F. M. Kifetew and P. Tonella. 2018. **Automated Test Case Generation as a Many-Objective Optimisation Problem with Dynamic Selection of the Targets**. IEEE Transactions on Software Engineering. vol. 44, no. 2, pp. 122–158. DOI: https://doi.org/10.1109/TSE.2017.2663435

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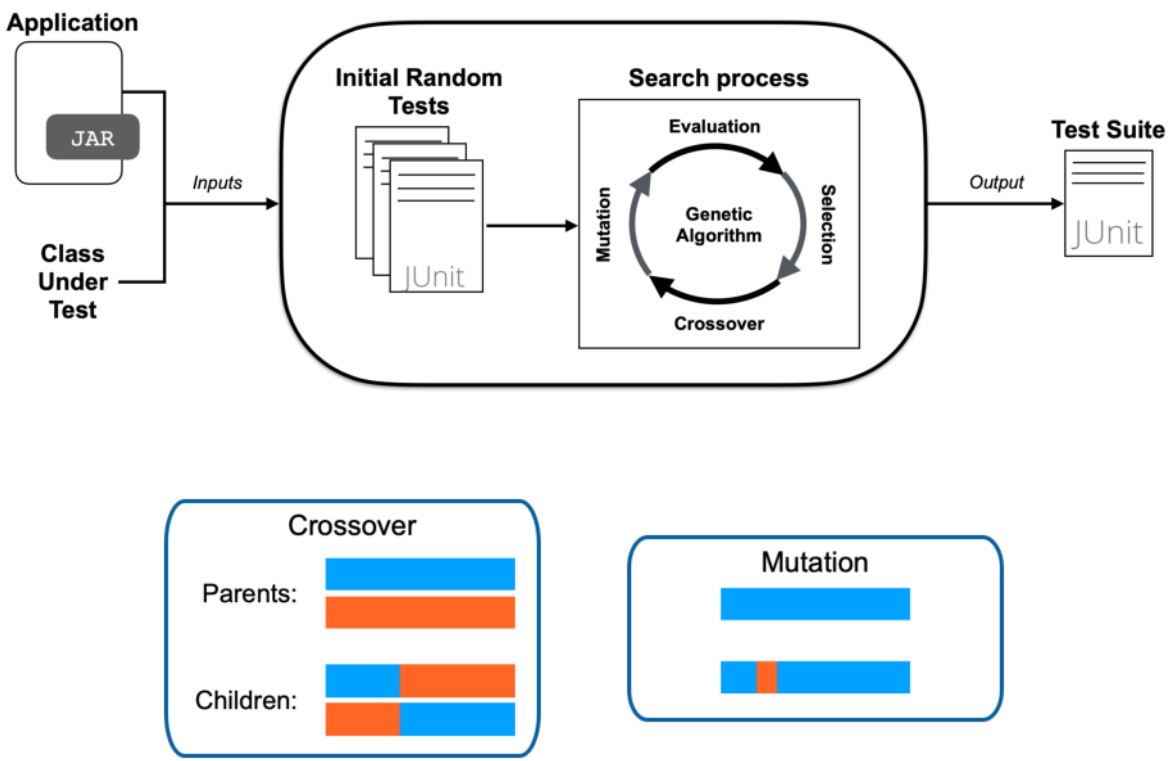
Context

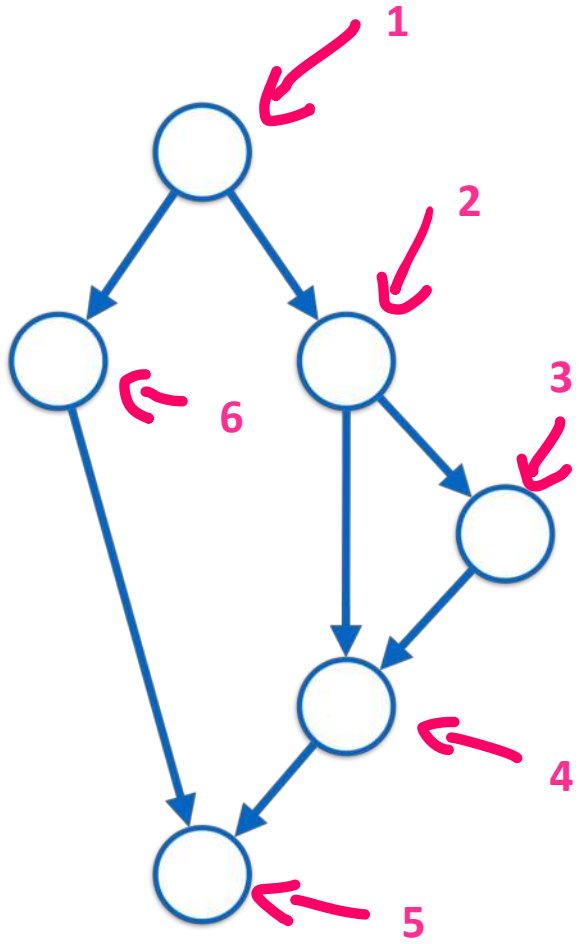
- White-box automated test case generation for Java
- Using genetic algorithms
- Generate a test suite maximizing a given set of coverage criterion (e.g., line coverage, branch coverage, weak mutation coverage, etc.)



Context

- Building test sequences and test input data covering specific objectives (like specific lines or branches)
- Existing algorithms consider objectives one at a time or aggregate all objectives into one fitness function

```
if […] then
  […]
else
  […]
  if […] then
    […]
  end
  […]
end
```



Context

- Example of existing algorithm: **MOSA**, a many-objective sorting algorithm
- In which all not-yet-satisfied objectives are considered in parallel

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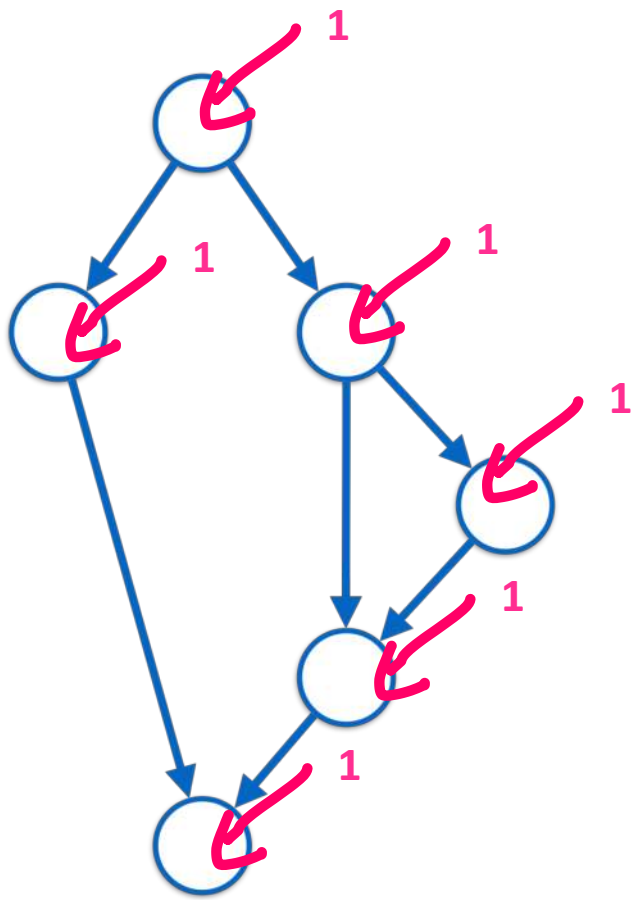
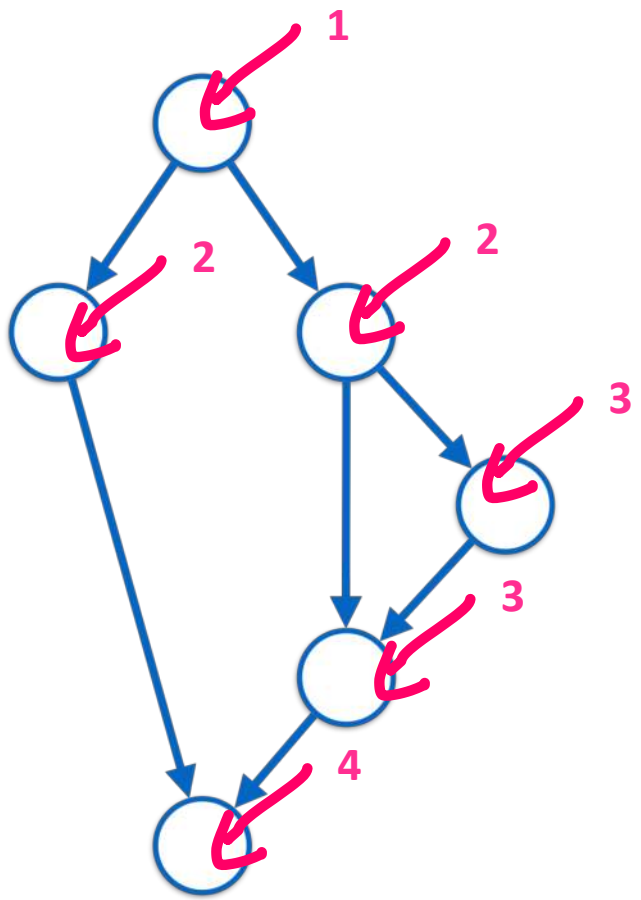
if […] then
  […]
else
  […]
  if […] then
    […]
  end
  […]
end
```



Objective

- Addressing the test generation problem using **DynaMOSA**, a dynamic many-objective sorting algorithm
- In which all **reachable** not-yet-satisfied objectives are considered in parallel

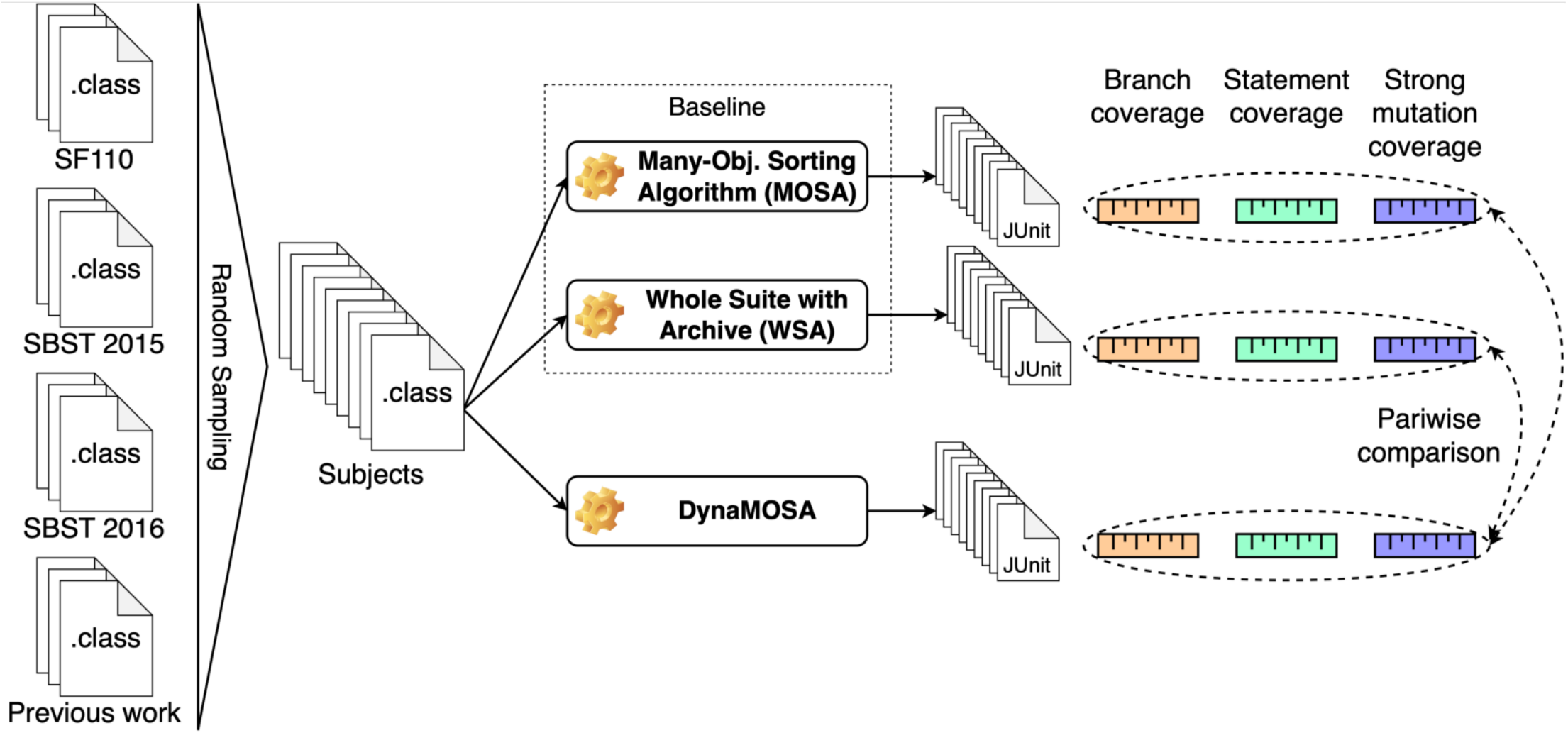
```
if […] then
  […]
else
  […]
  if […] then
    […]
  end
  […]
end
```



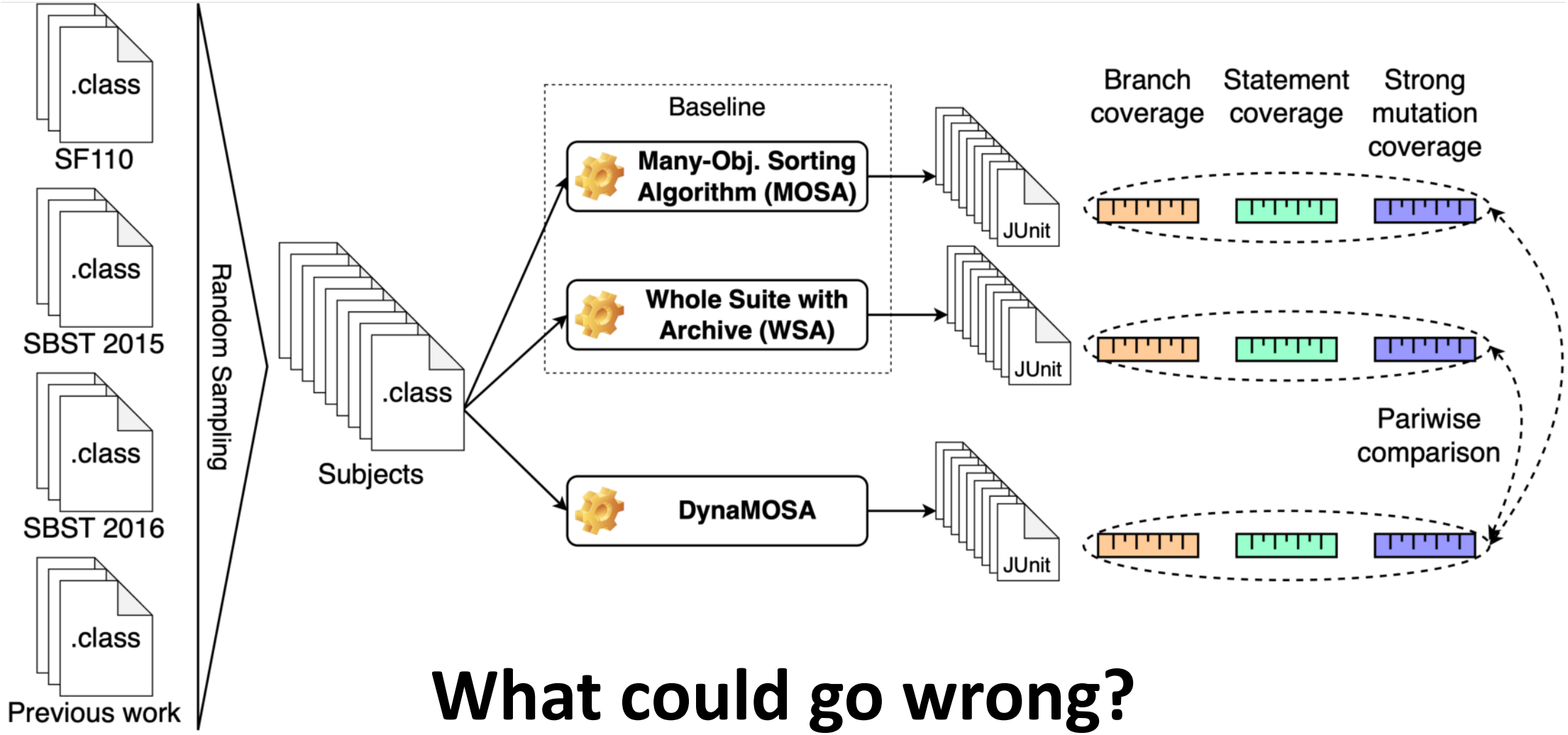
Research questions

- ***RQ$_1$*** *How does DynaMOSA perform compared to alternative approaches on branch coverage?*
- ***RQ$_2$*** *How does DynaMOSA perform compared to alternative approaches on statement coverage?*
- ***RQ$_3$*** *How does DynaMOSA perform compared to alternative approaches on strong mutation coverage?*
- ***RQ$_4$*** *How do preference criterion and Pareto dominance affect the effectiveness of DynaMOSA?*
  - ***RQ$_{4.1}$*** *What is the effectiveness of DynaMOSA's Pareto-based ranking alone, when applied without the preference criterion?*
  - ***RQ$_{4.2}$*** *What is the effectiveness of DynaMOSA's preference criterion alone, when applied without the Pareto-based ranking?*

Method: Optimization Studies in SE

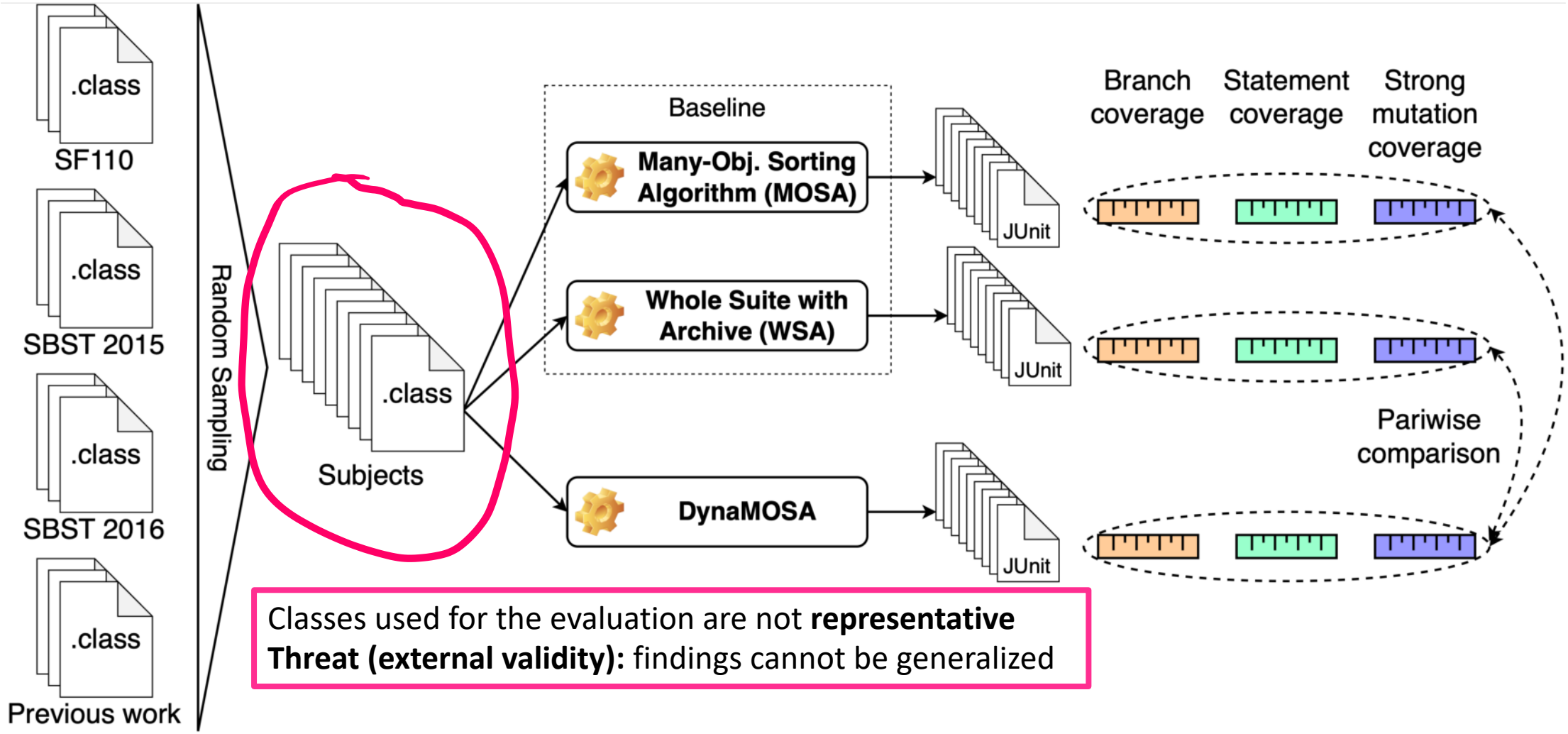


Method: Optimization Studies in SE



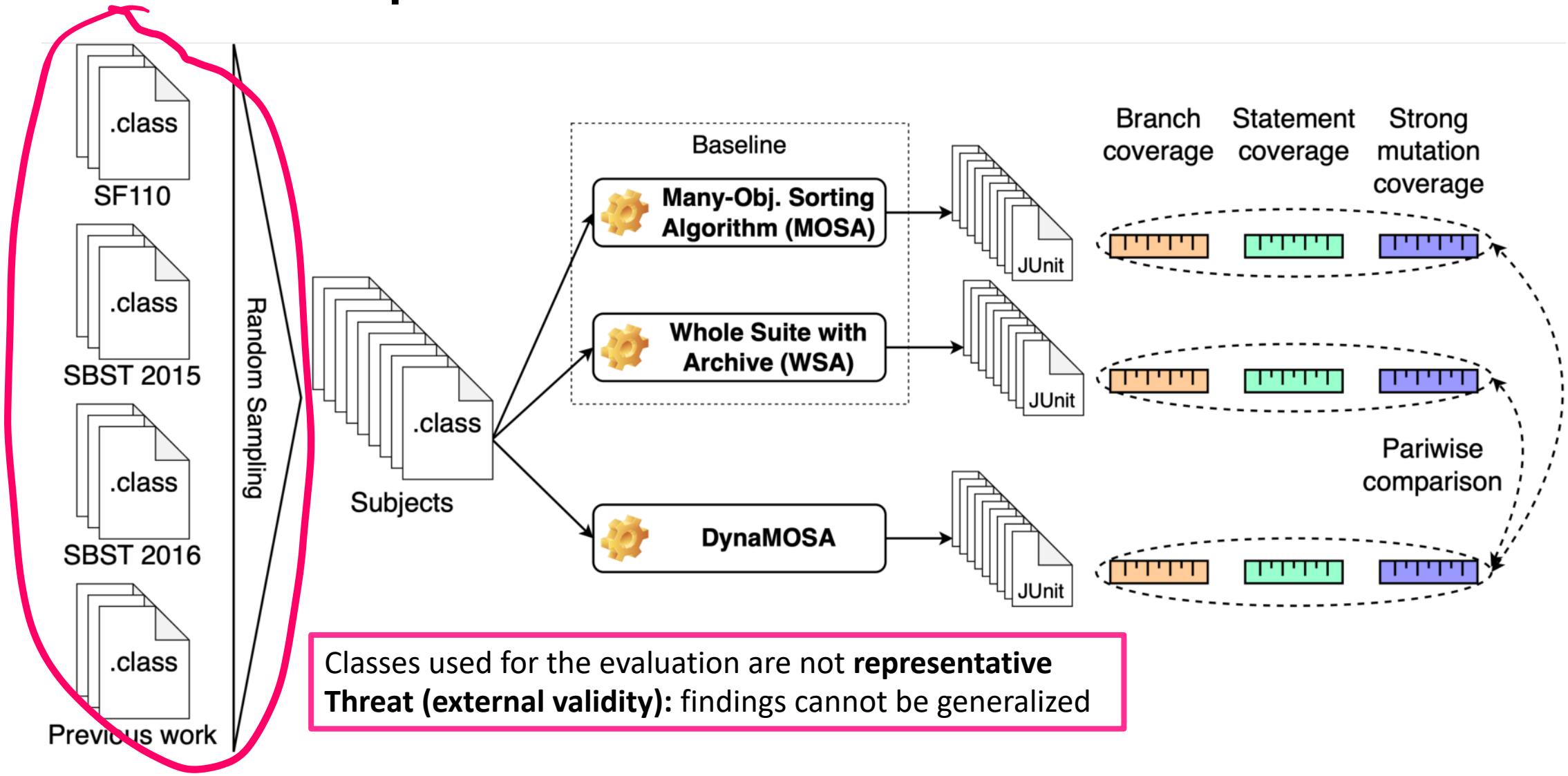
What could go wrong?

Method: Optimization Studies in SE



Classes used for the evaluation are not **representative**
Threat (external validity): findings cannot be generalized

Method: Optimization Studies in SE



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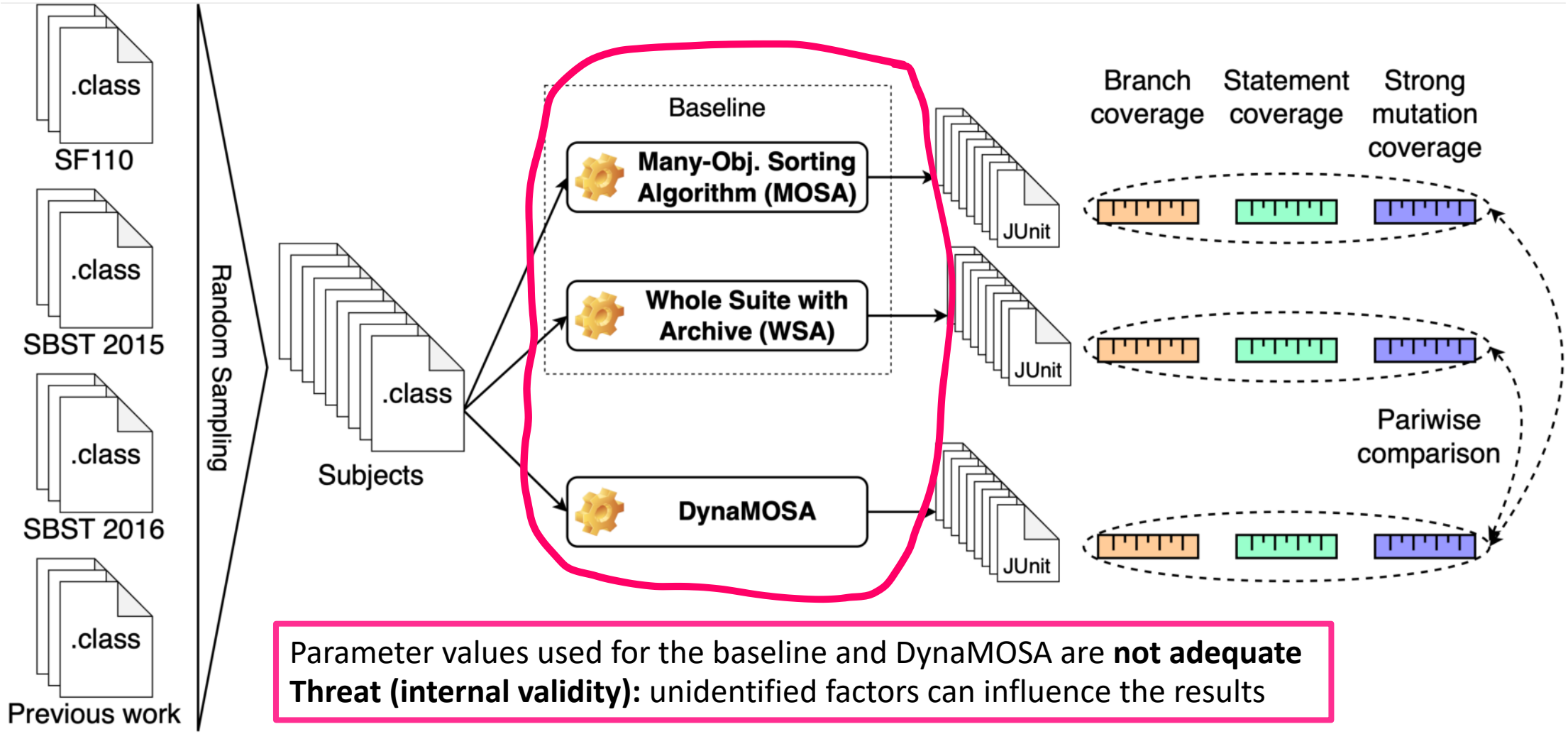
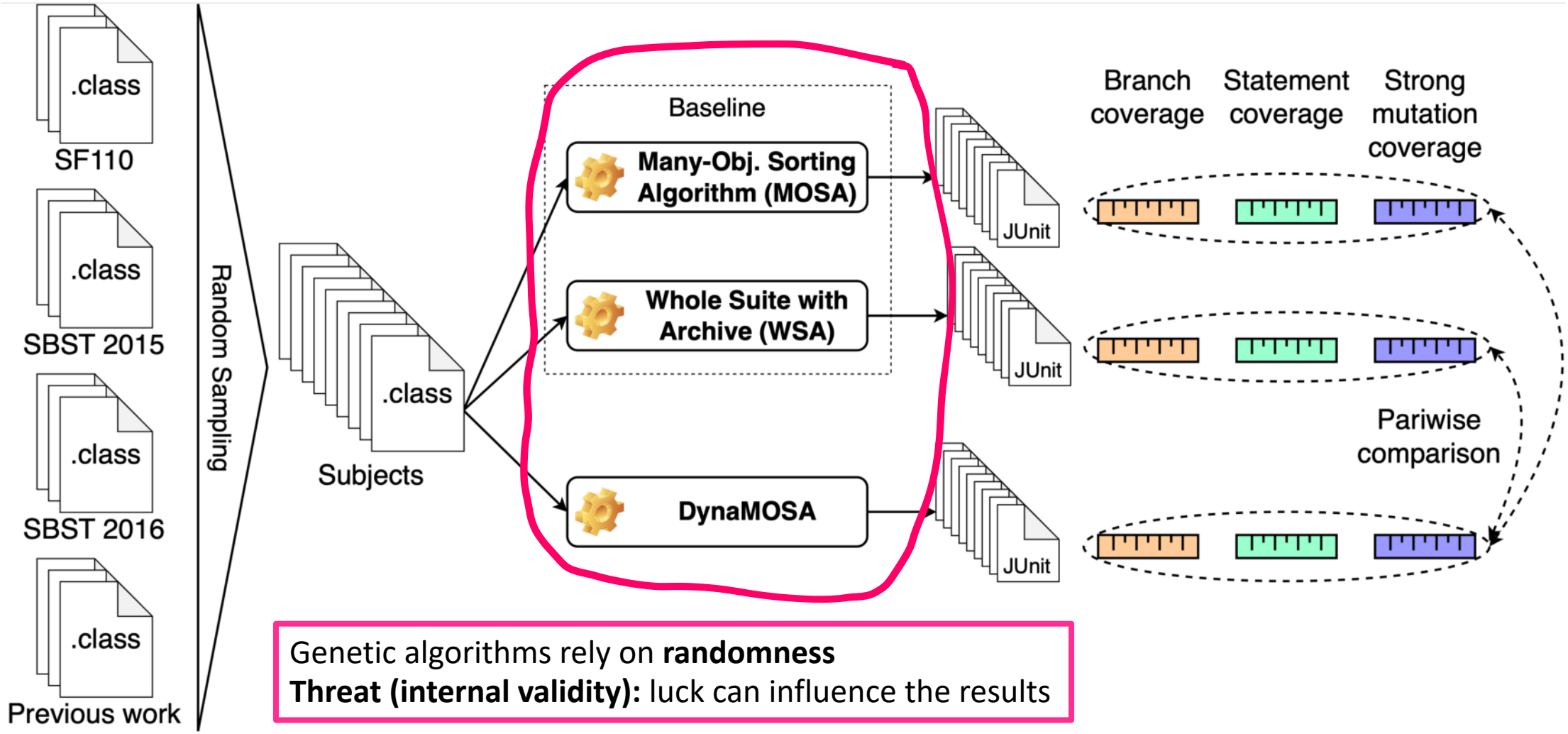
Method: Optimization Studies in SE



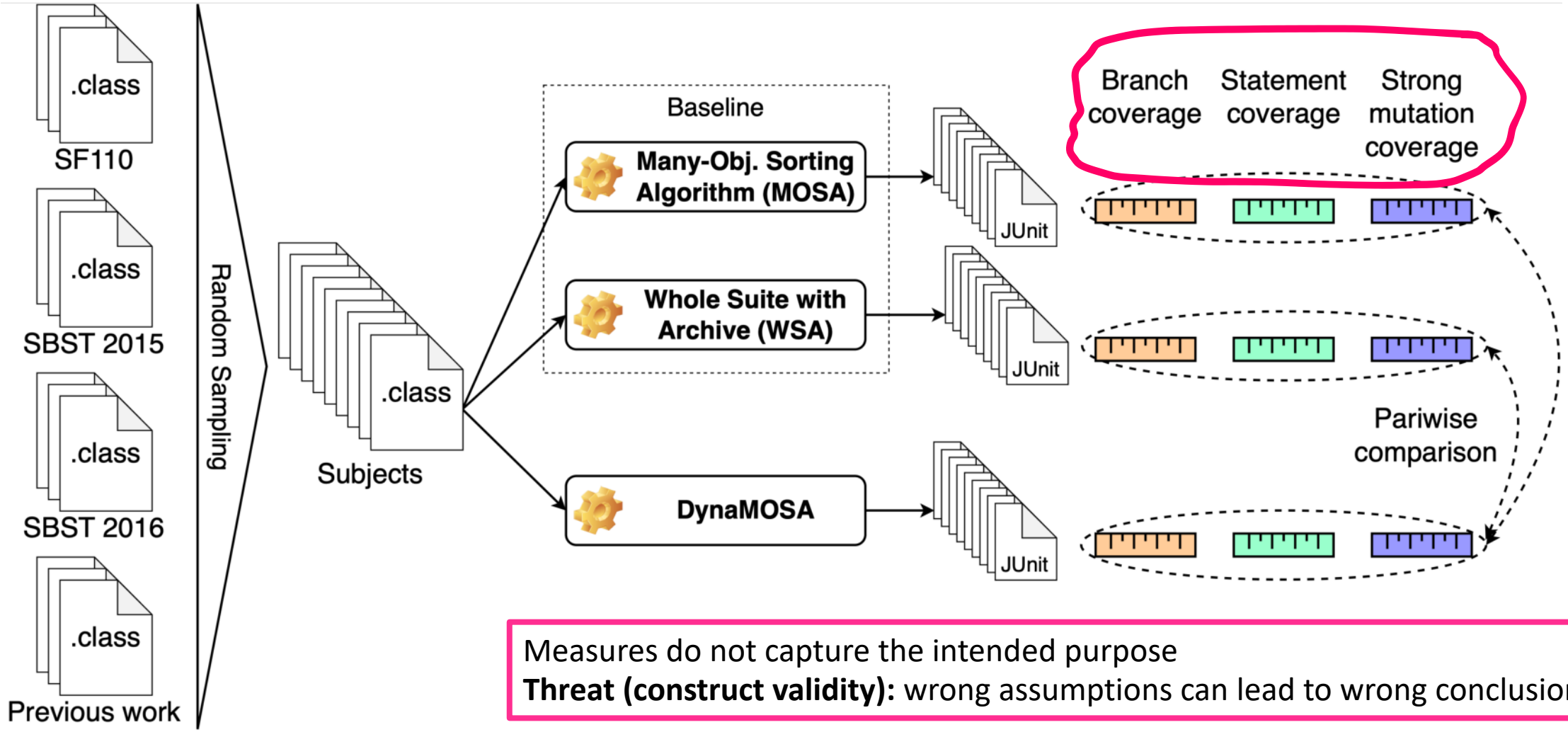
Parameter values used for the baseline and DynaMOSA are **not adequate**
Threat (internal validity): unidentified factors can influence the results

Method: Optimization Studies in SE



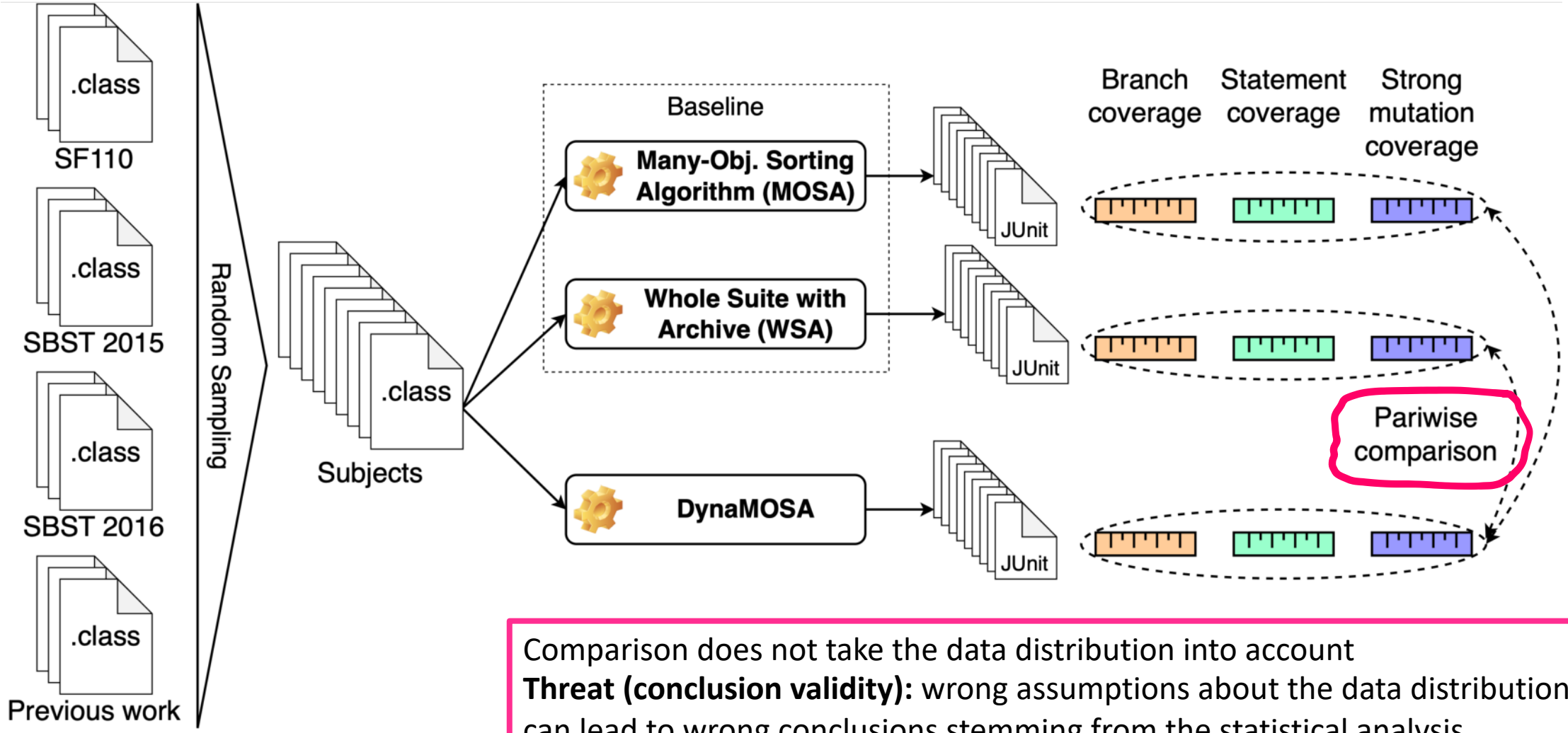
Genetic algorithms rely on **randomness**
Threat (internal validity): luck can influence the results

Method: Optimization Studies in SE



Measures do not capture the intended purpose
Threat (construct validity): wrong assumptions can lead to wrong conclusions

Method: Optimization Studies in SE



Comparison does not take the data distribution into account
Threat (conclusion validity): wrong assumptions about the data distribution can lead to wrong conclusions stemming from the statistical analysis

Research method: How?

It all starts with an onion

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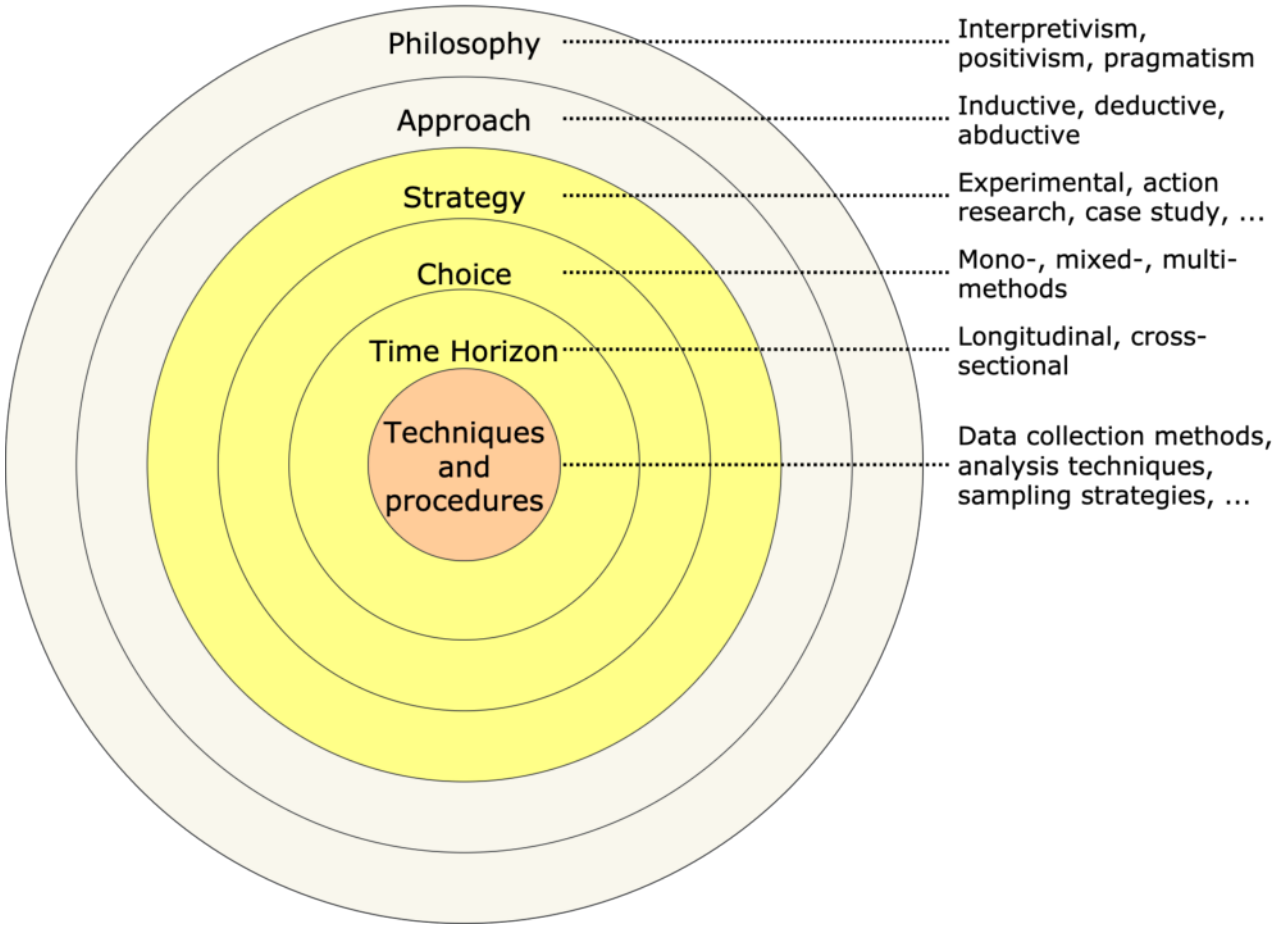
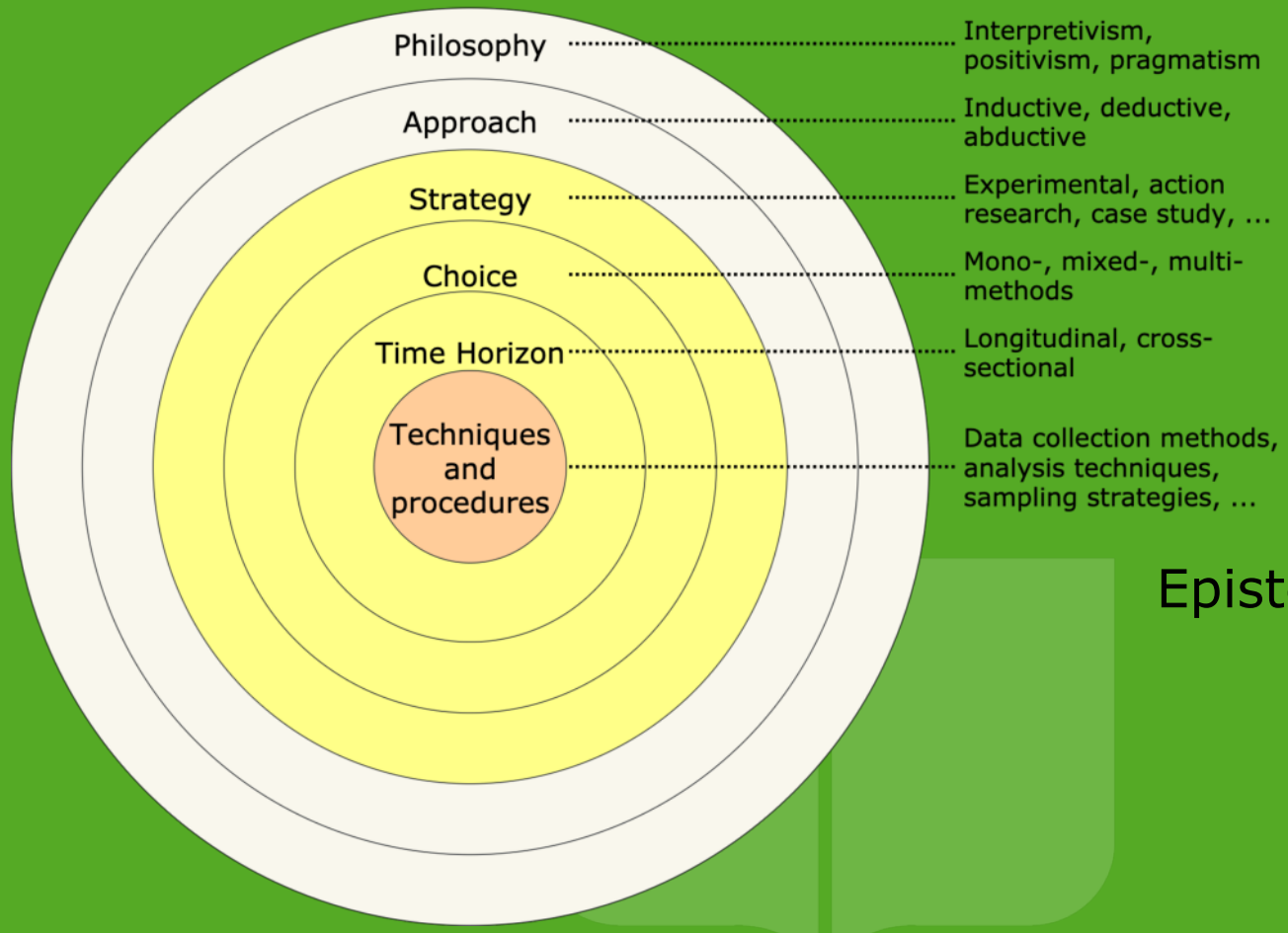
Methodology

- Study of research methods
- Offers a framework for scientific research
- Helps defining a research method
- Standardize practices to increase confidence in the results and reduce threats to the validity
- Enables reproducibility of research in different contexts
- Supports the emergence of new research methods

Onion model (Saunders et al.)



Saunders, Mark N. K.; Bristow, Alexandra; Thornhill, Adrian and Lewis, Philip (2019). Understanding research philosophy and approaches to theory development. In: Research Methods for Business Students, 8th edition. Harlow: Pearson Education, pp. 128–171.



Philosophy

Epistemology - How is knowledge obtained?

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Philosophy: Positivism

- Knowledge is objective and can be obtained by hypothesis testing
- Rigorous, controllable method
- No interpretation of the observations
- The “why” is irrelevant
- Quantitative empirical methods

Auguste Comte (1798 - 1857)

Philosophy: Interpretivism

- Social phenomenon cannot be studied with a positivist stance
  - Studying of human being by human being cannot be objective
- Researchers have their own perceptions and interpretations
  - Knowledge is built from interpretation
- Qualitative methods

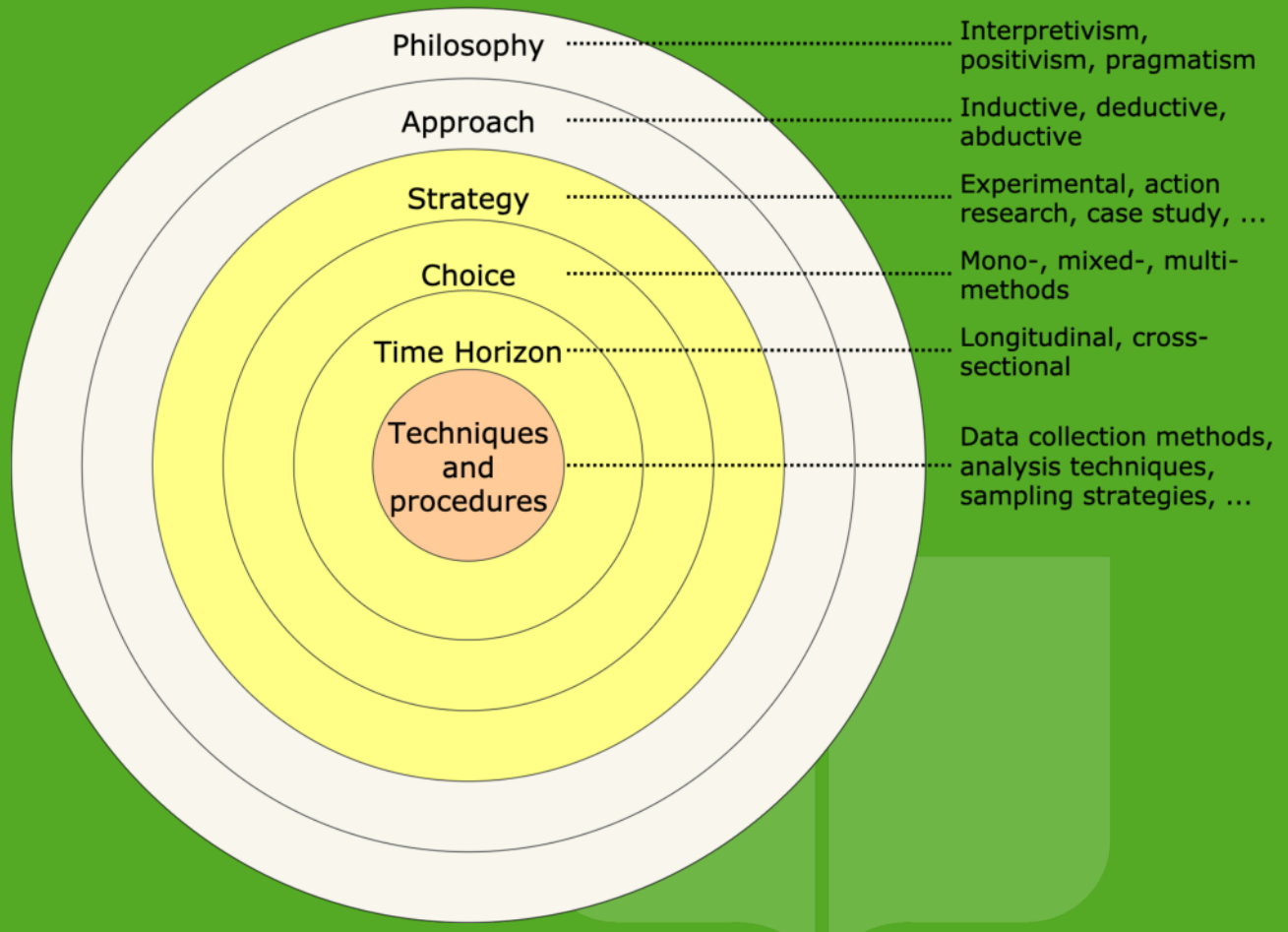
"What is real? How do you define 'real'? If you're talking about what you can feel, what you can smell, what you can taste and see, then 'real' is simply electrical signals interpreted by your brain."

Morpheus, The Matrix (1999)

Philosophy: Pragmatism

- Multiple stances, best viewed in terms of their practical usages
- The pragmatic maxim: "*Consider the practical effects of the objects of your conception. Then, your conception of those effects is the whole of your conception of the object.*"
- A scientific concept or theory should be evaluated by how effectively it explains and predicts phenomena, as opposed to how accurately it describes objective reality

Charles Peirce (1839 - 1914)



Approach

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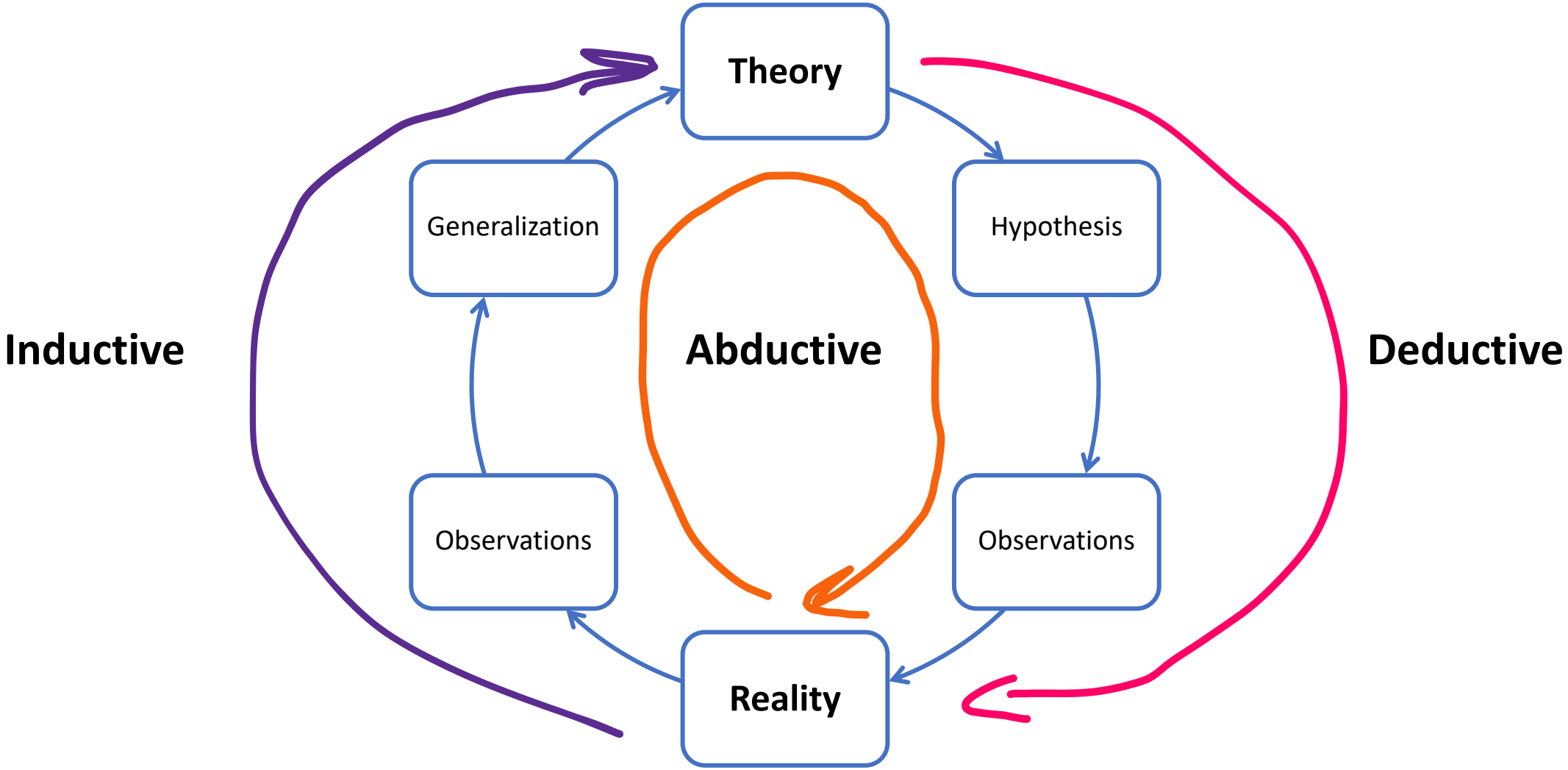
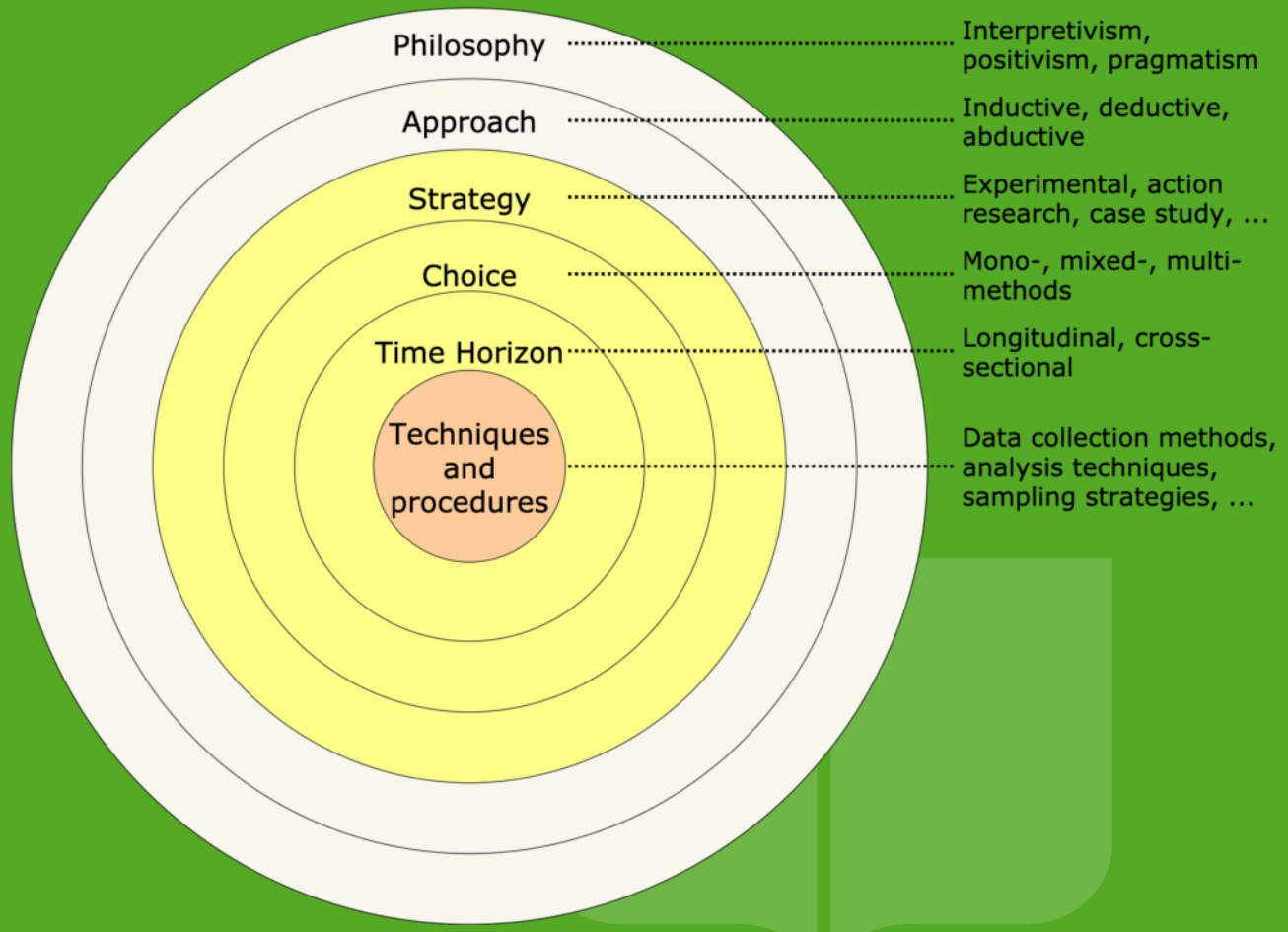
Deductive, Inductive, or Abductive





Strategy

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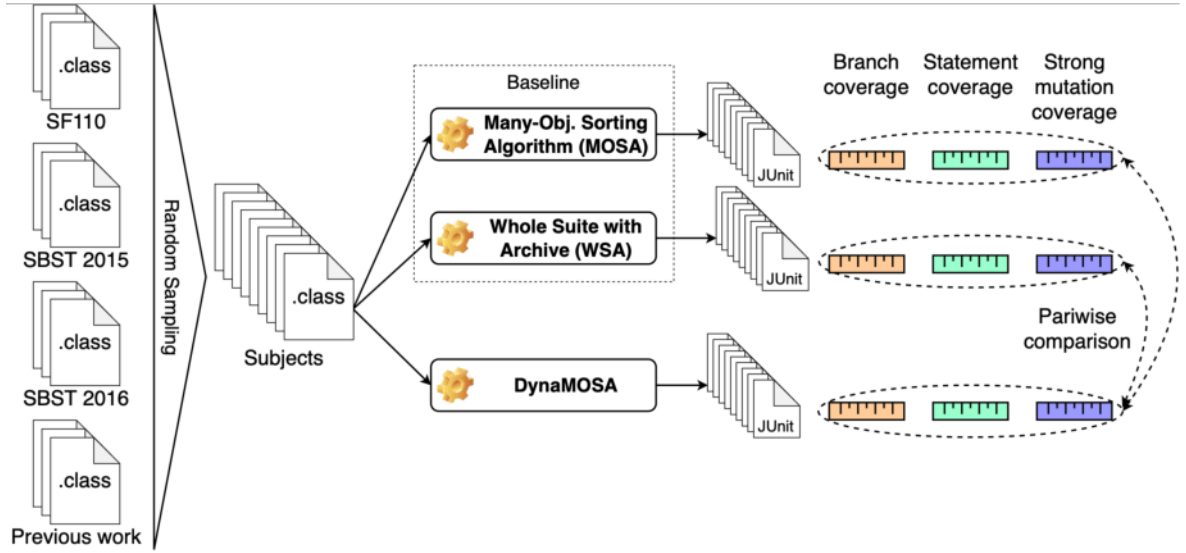
Experiment

- Measure the effects of an intervention on a population
  - Independent variables
    - Explanatory/Predictor
  - Dependent variables
    - Outcome to measure
- Controlled environment (in lab)



Survey

- Collect information from a subgroup of a population
- Sampling strategy
  - Random
  - Opportunistic
  - Stratified
  - …
- Used to
  - explore (identify important constructs),
  - describe (tell what is happening),
  - or explain (test theory)

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Guide en matière d'enquêtes par voie électronique

Préambule : Ce que l'on entend par enquêtes

Dans le présent guide sont visées les enquêtes comprises comme un processus électronique permettant de recueillir auprès de personnes, de firmes ou d'organismes (préalablement sélectionnées) des données qualitatives ou quantitatives. Il peut s'agir tant d'enquêtes réalisées par les services institutionnels l'UNamur que d'enquêtes réalisées dans le cadre de d'études ou de recherches de membres du personnel ou d'étudiants (dans le cadre d'un mémoire, par exemple).

Étape 1 : Objectif de l'enquête

Avant de commencer à choisir un outil, il est nécessaire de définir clairement deux éléments liés directement à votre enquête :

- le contexte dans lequel cette enquête va être menée ; quelle est la situation qui vous a amené à chercher ces données ? Quelle est la question (d'études ou de recherches à laquelle vous cherchez à répondre ?
- l'objectif de cette enquête ; quelle information souhaitez-vous obtenir sur base de l'analyse de ces données et comment allez-vous utiliser cette information ?

Prenez le temps de vous poser ces deux questions et de pouvoir y répondre clairement avant de vous lancer dans la création d'une enquête. Une fois cela mis au clair, vous aurez ainsi une ligne de conduite qui vous aidera dans la conception du questionnaire et dans l'analyse des données obtenues.

Étape 2 : choix de l'outil d'enquête

Le recours à l'un ou l'autre outil peut tout d'abord dépendre du type d'enquête (qualitative ou quantitative, à questions ouvertes ou fermées).

Critère du type d'enquête

Une enquête quantitative donne des résultats quantifiables, c'est-à-dire des données mesurables qui peuvent être converties en chiffres. On parle ici d'une étude descriptive dont l'objectif est de décrire et recueillir des données.

L'analyse de ces données permettent de mesurer l'intérêt d'un public pour un produit, la satisfaction d'un usager, de valider une hypothèse développée …

L'enquête qualitative a pour objectif de comprendre ou d'expliquer un comportement, des motivations et des caractéristiques. Elle collecte des impressions, opinions et avis sous forme verbale ou orale plutôt que des données chiffrées. Les informations devront être analysées de façon interprétative et subjective. Cette analyse permet de comprendre ou expliquer un comportement spécifique, des motivations ou attitudes des personnes sondées. Dans ce contexte

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Case study

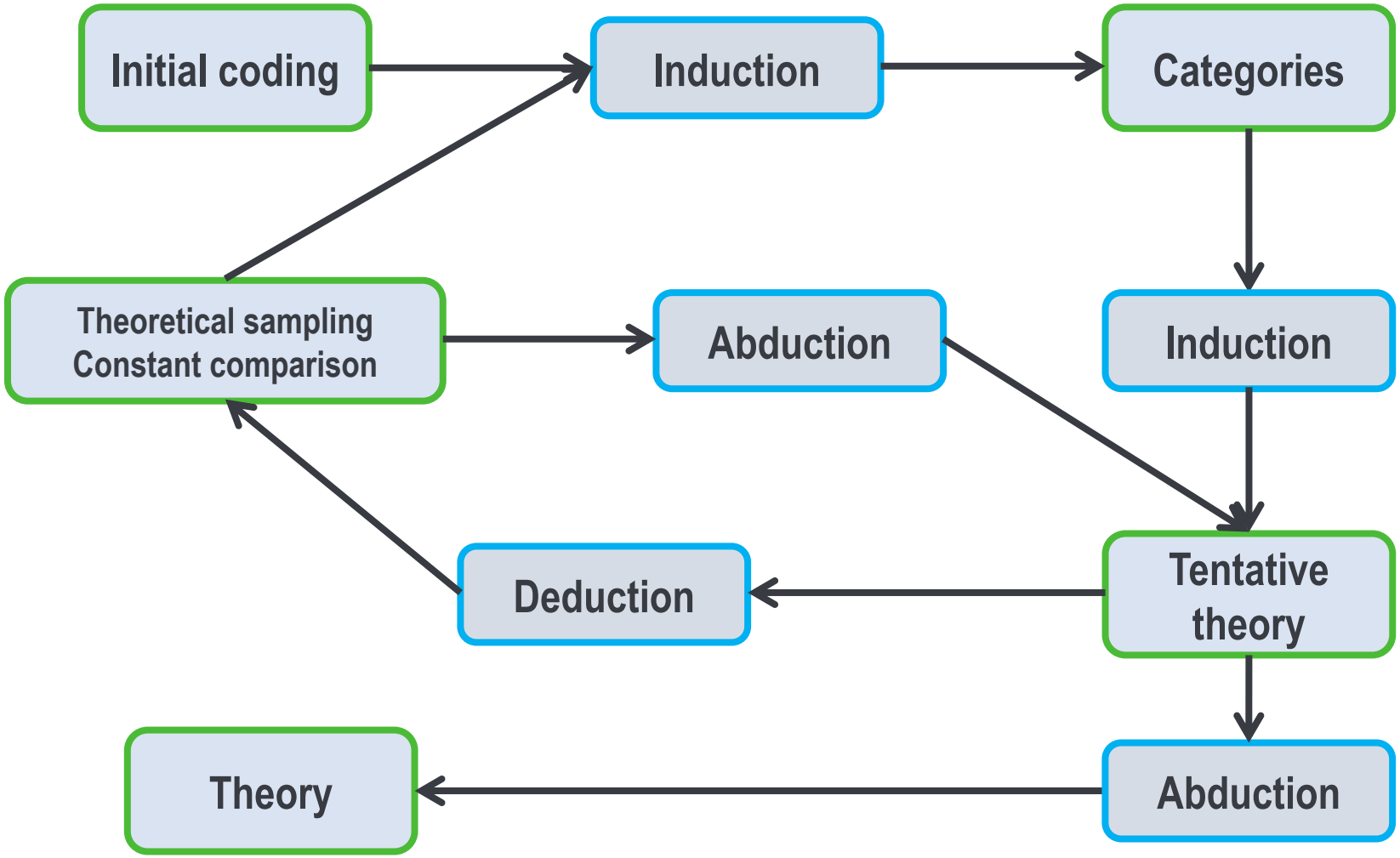
- Focus on a phenomenon, an individual, a group, or an organization
- Gain deep understanding
- Used to
  - explore (identify important constructs),
  - describe (tell what is happening),
  - or explain (test theory)
- Relies on several techniques to collect data at different points in time

Action research

- Problem-driven
- Introduce an action in an environment and study its effects
- Aims at contributing to
  - research (by expanding scientific knowledge)
  - and practice (by solving a practical problem)
- Iterative

Grounded theory

- Develop a new theory from (qualitative) data
- Iterative process
  - Identify important categories and relationships
  - Collect more data until saturation



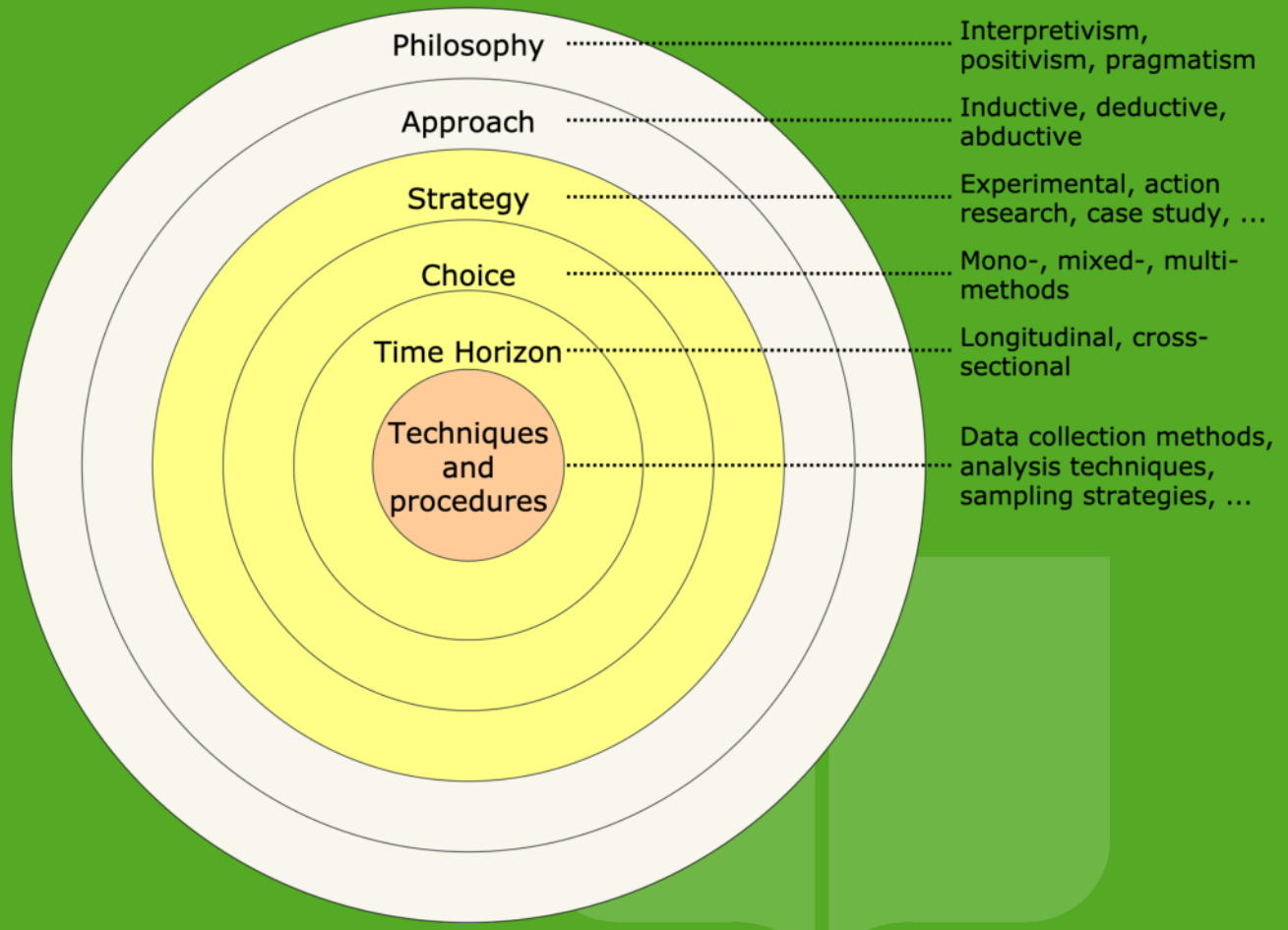
Ethnography

- Study an individual, a group, an organization in a detached way
- No intervention
- Controlled
  - Define tools and observation protocols
- Or uncontrolled

Archival research

- Examine existing documents (historical archives) to draw conclusions
  - Project reports
  - Source code history
  - Issue tracking systems
  - Build history
  - …
- Comparable to systematic literature reviewing

https://conf.researchr.org/series/msr



Choice

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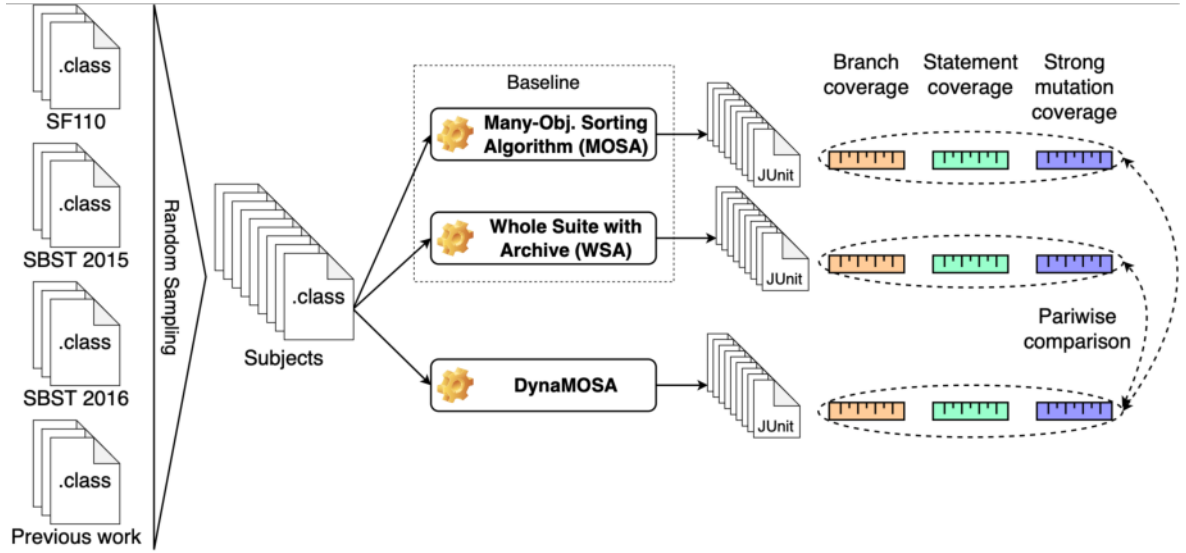
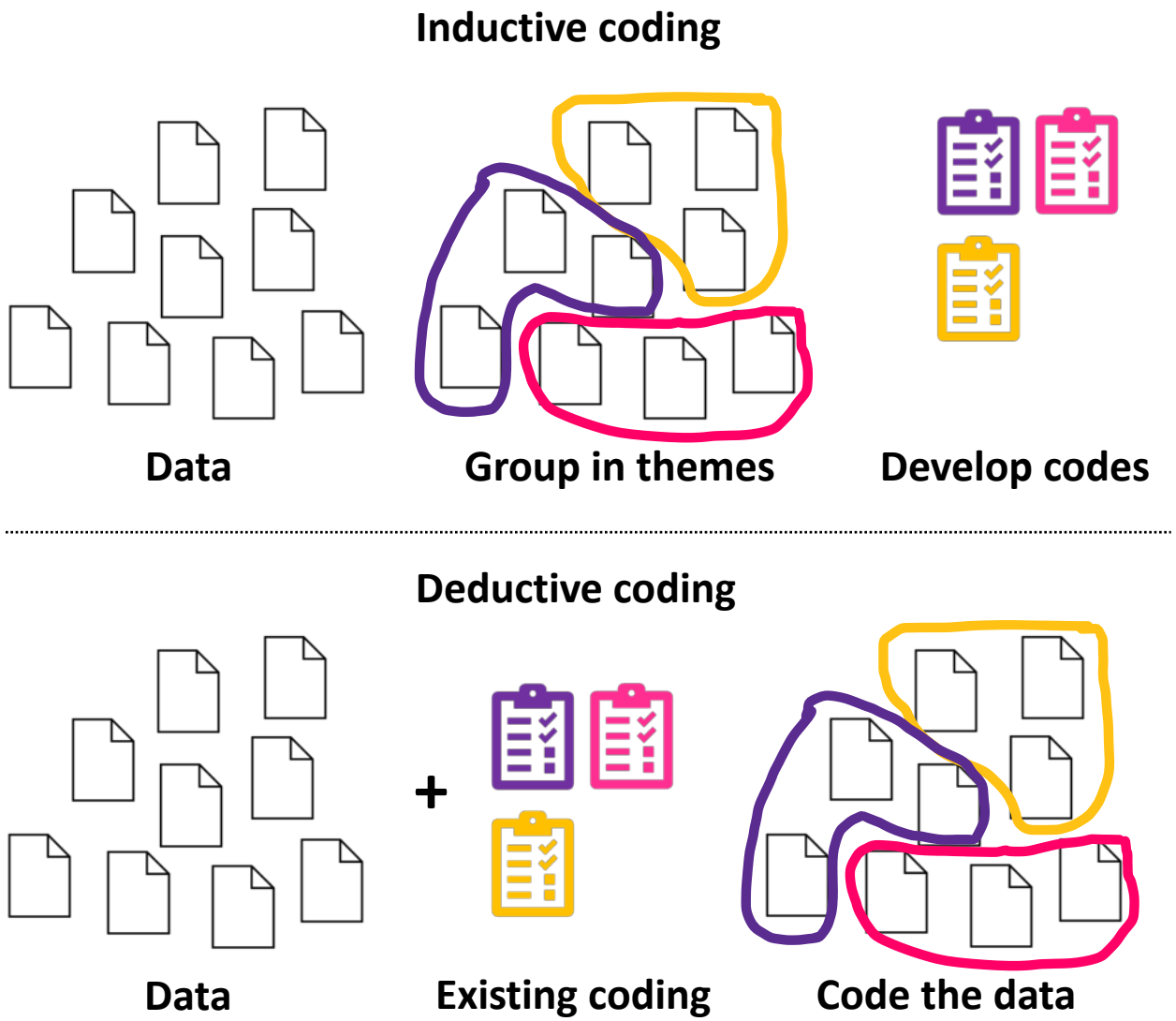
Quantitative methods

- Focused on quantitative data (i.e., numbers)
- Highly-structured tools used to collect data
  - Questionnaires (close-ended questions)
  - Well-defined metrics
- Large sample size

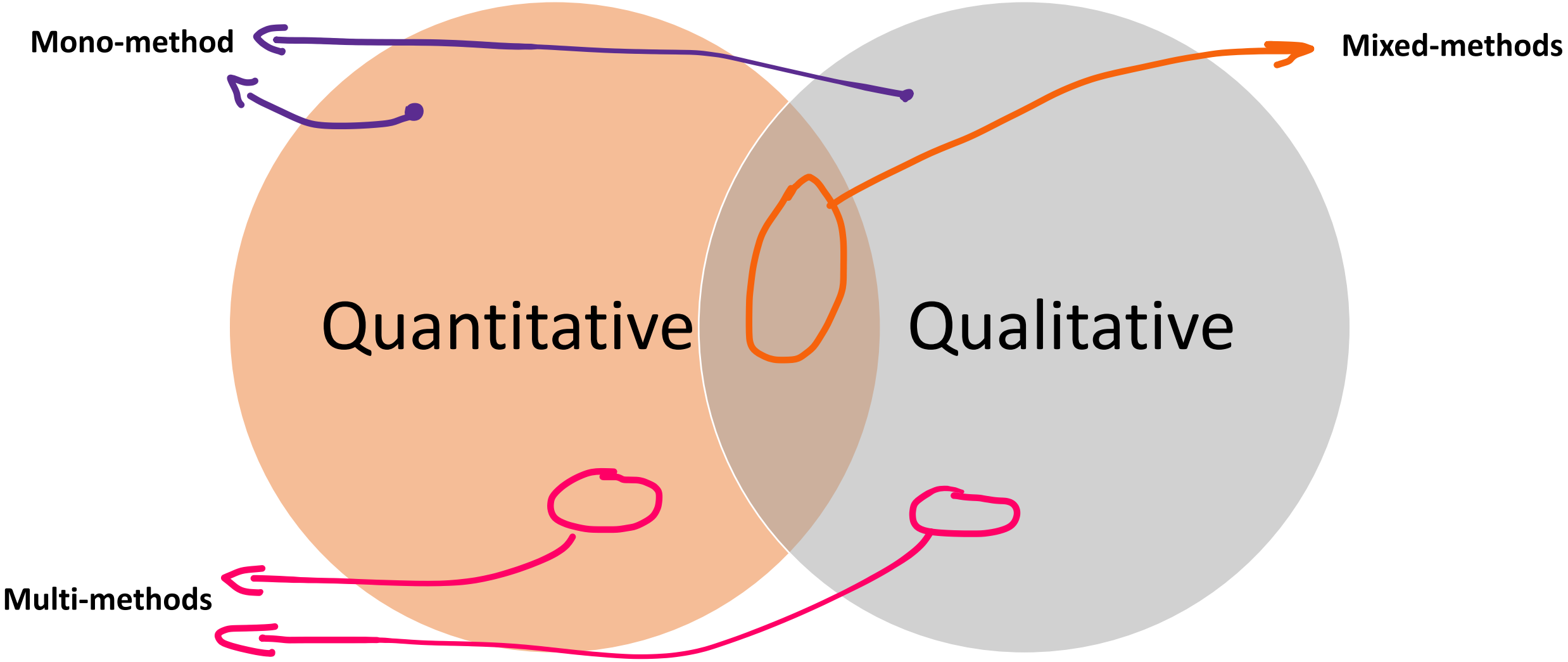


Qualitative methods

- Focused on data that are textual, visual, audio, …
- Understand concepts, opinions, or experiences
- Gather in-depth insights
- Data collection
  - Observations
  - Interviews
  - Texts, images, videos, etc.
- Relies on coding

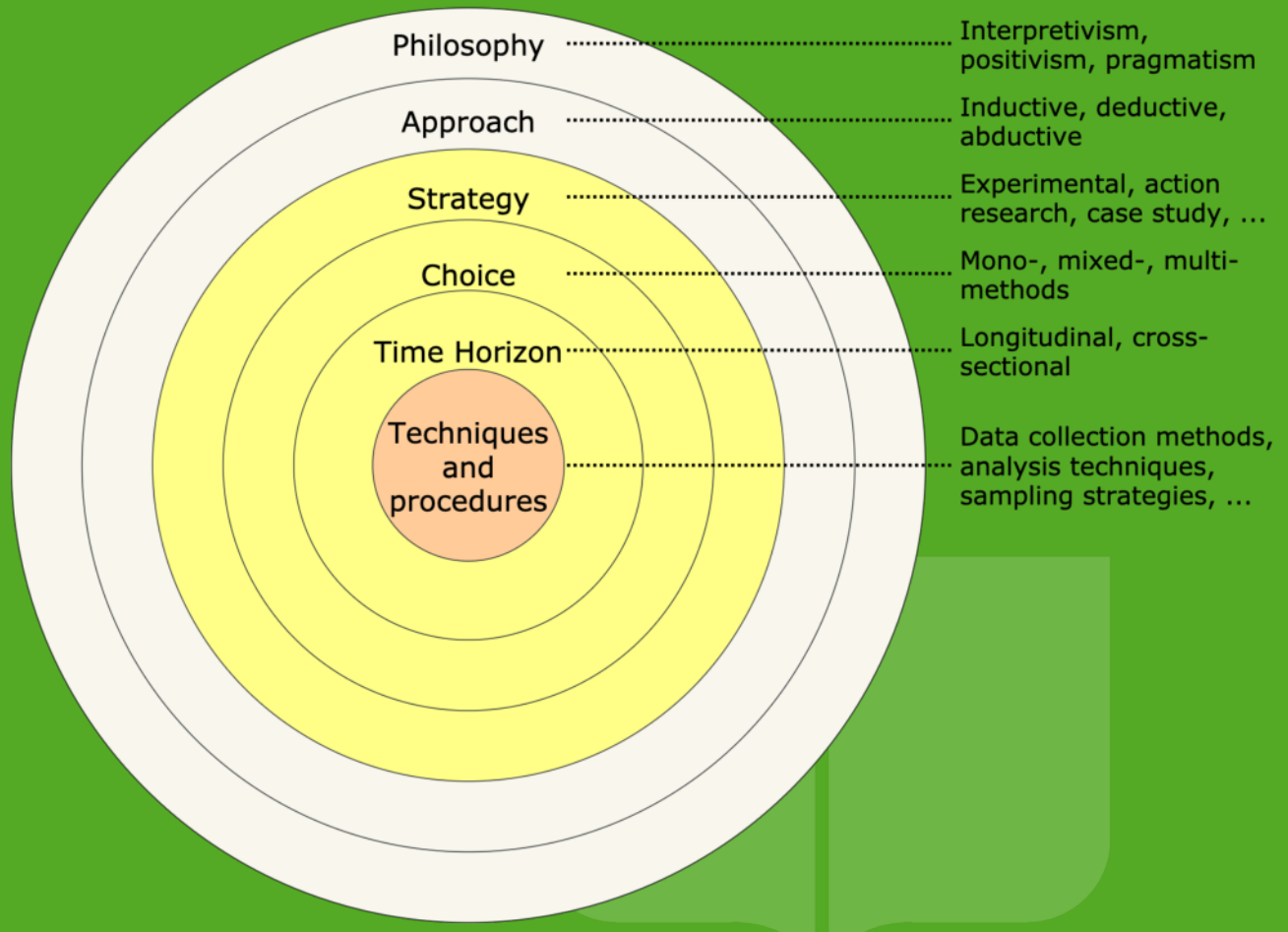


Methods


Mono-method
Mixed-methods
Quantitative
Qualitative
Multi-methods

Methods

- Triangulation
  - Check whether results from several methods converge
- Complementarity
  - Use different methods for different parts of the research
- Initiation
  - Discover contradictions to reframe the research questions
- Development
  - Use the results of one method to develop the other
- Expansion
  - Expand the breadth and range of research



Time Horizon

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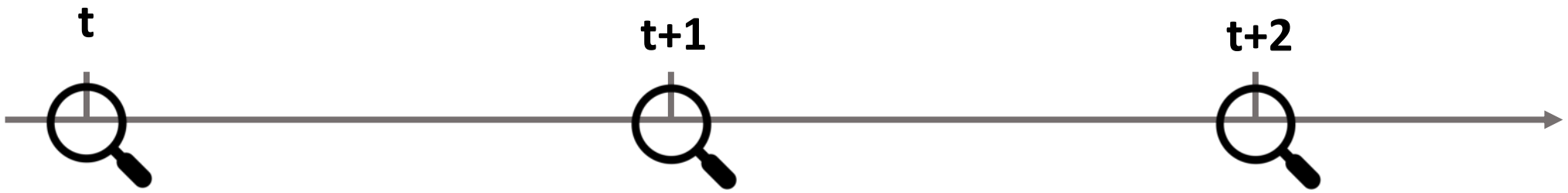
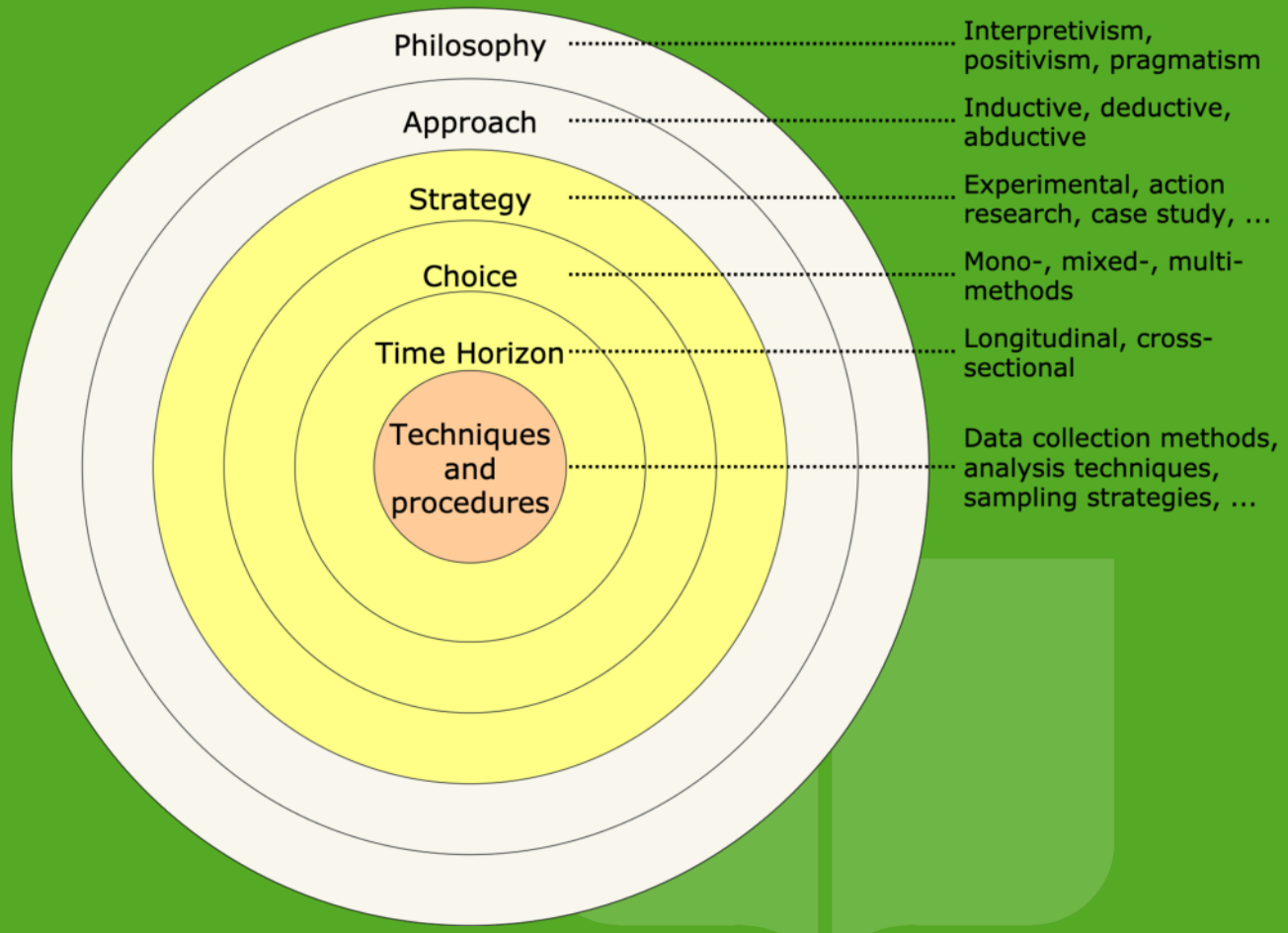
Cross-sectional

- Measure a construct once

Longitudinal

- Measure the same construct at several time intervals
  - Minutes, hours, days, years
- Sampling
  - Panel: sample of people
  - Cohort: sample of people having a common feature
  - Retrospective: based on historical records





Techniques and procedures

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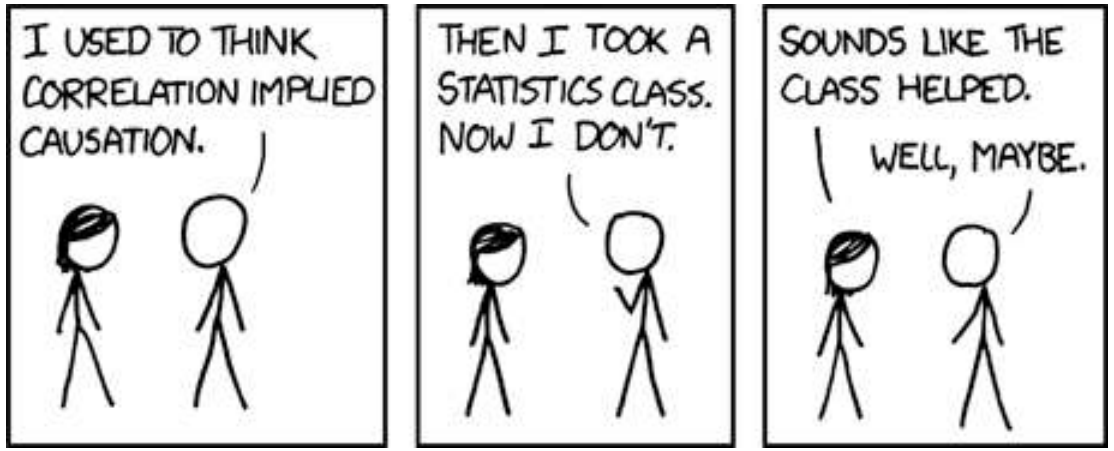
Data collection

- Which data source?
- When is it collected?
- How is it collected?
  - Questionnaire
    - Open-ended or close-ended
    - Existing or new
  - Interviews
    - Interview plan (Structured, semi-structured, unstructured)
  - Automated tools
    - Tool + metrics + configuration
  - …

Data analysis

- Coding scheme
  - Existing scheme
  - New scheme
- Statistical analysis
  - Description of the procedure
  - Checks on the data distribution
  - Check for p-values
- Cross-validation
- …



Recommended reading

https://onlinelibrary.wiley.com/doi/10.1002/stvr.1486

SOFTWARE TESTING, VERIFICATION AND RELIABILITY
Softw. Test. Verif. Reliab. 2014; **24**:219–250
Published online 6 November 2012 in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/stvr.1486

A Hitchhiker's guide to statistical tests for assessing randomized algorithms in software engineering‡

Andrea Arcuri [1,*,†] and Lionel Briand [2]

[1] *Simula Research Laboratory, P.O. Box 134, Lysaker, Norway*
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SUMMARY

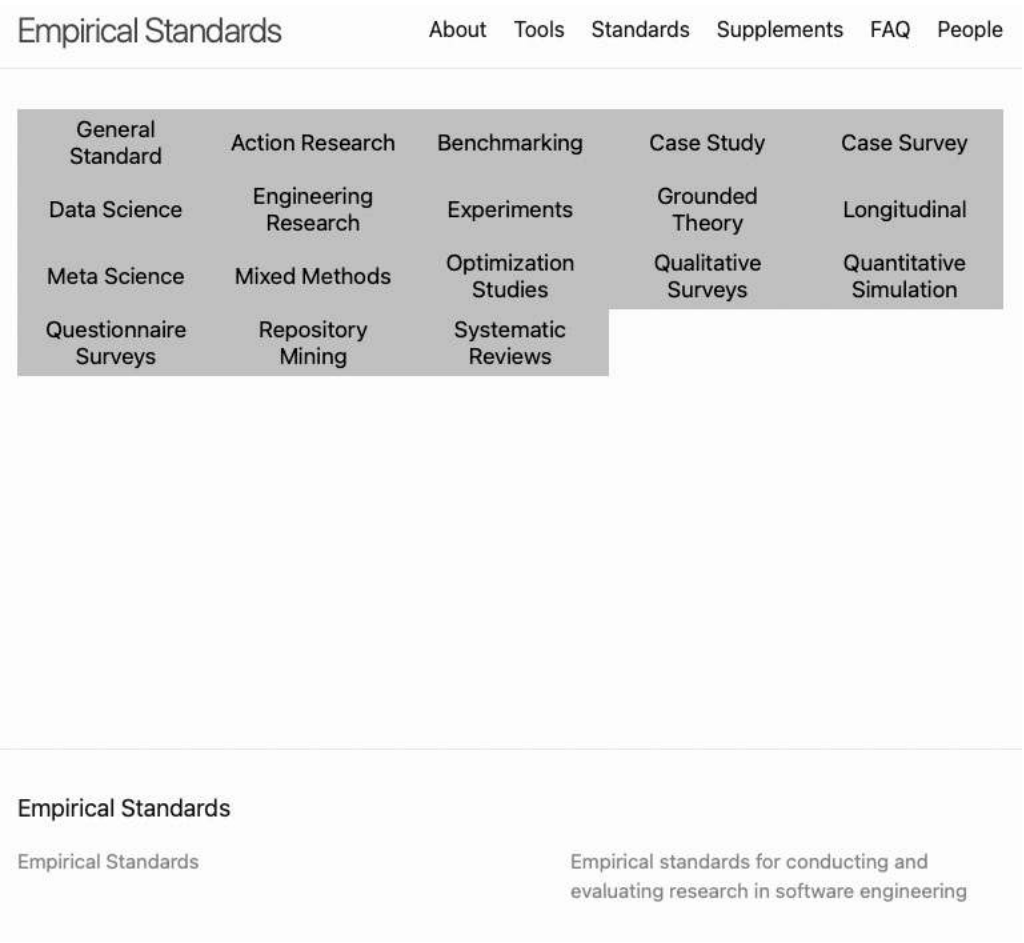
Randomized algorithms are widely used to address many types of software engineering problems, especially in the area of software verification and validation with a strong emphasis on test automation. However, randomized algorithms are affected by chance and so require the use of appropriate statistical tests to be properly analysed in a sound manner. This paper features a systematic review regarding recent publications in 2009 and 2010 showing that, overall, empirical analyses involving randomized algorithms in software engineering tend to not properly account for the random nature of these algorithms. Many of the novel techniques presented clearly appear promising, but the lack of soundness in their empirical evaluations casts unfortunate doubts on their actual usefulness. In software engineering, although there are guidelines on how to carry out empirical analyses involving human subjects, those guidelines are not directly and fully applicable to randomized algorithms. Furthermore, many of the textbooks on statistical analysis are written from the viewpoints of social and natural sciences, which present different challenges from randomized algorithms. To address the questionable overall quality of the empirical analyses reported in the systematic review, this paper provides guidelines on how to carry out and properly analyse randomized algorithms applied to solve software engineering tasks, with a particular focus on software testing, which is by far the most frequent application area of randomized algorithms within software engineering. Copyright © 2012 John Wiley & Sons, Ltd.

D4 Poster méthode de recherche

UNIVERSITÉ DE NAMUR

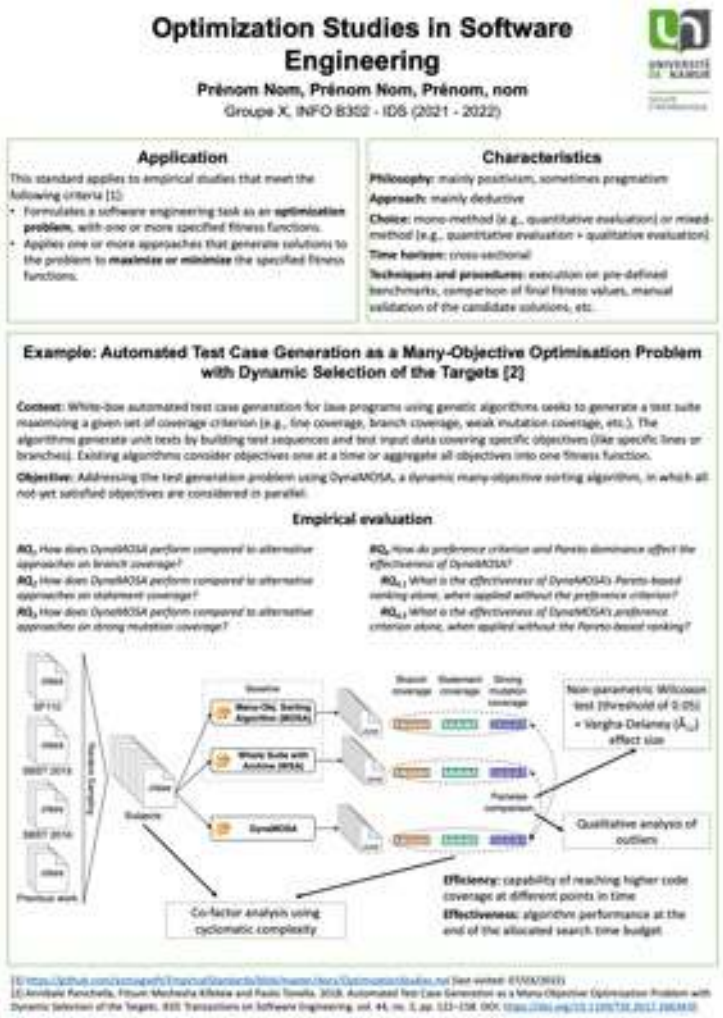
Empirical Standards (for SE research)

- https://acmsigsoft.github.io/EmpiricalStandards/docs/
- Standardizes peer-reviewing
- Authors declare their research methods
- Reviewers receive a specific checklist
- Goal: reduce review time and biases



D4 Poster méthode de recherche

- Choisir une méthode de recherche parmi celles disponibles
- Présenter les caractéristique générale de la méthode
- Choisir un exemple (*Exemplar*)
- Présenter brièvement le sujet et (surtout !) le protocole expérimental développé



Considérations pratiques

- Choisir de préférence une méthode susceptible de répondre à la question de recherche décrite dans le déliverable D3
- Deux groupes ne peuvent pas choisir le même exemple (*Exemplar*)
  - Annoncer la méthode et l'exemple choisis sur Teams (canal General, en réponse au message qui sera posté)
  - Attribution faite sur base du premier arrivé, premier servi
- Respecter le format du template et envoyer le poster au format PDF (uniquement !)